Framework for Blockchain-Based Community-Assisted Privacy- Preserving Reporting Service for Local Governance

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Abstract

Local governments in developing countries often struggle to address public issues efficiently. These struggles are mainly due to limited transparency, centralized data management, and insufficient citizen participation. E-governance and crowd-sourced reporting systems are designed to enhance public engagement. But their effectiveness is limited by issues related to data integrity, user anonymity, content reliability, and trustworthiness.As a solution to the above challenges, we propose a blockchain-based, community-assisted, privacy-preserving reporting framework for local governance. The main objective of the project is to design and prototype a permissioned blockchain-based reporting system. This system will enable citizens to anonymously submit public issue reports and participate in opinion polling. Also, our system will ensure data immutability and transparency. To manage reporting workflows, issue tracking, and community voting in a secure and tamper-resistant manner, the blockchain will use Proof-of-Authority (PoA) mechanism and smart contracts. User identity is decoupled from the report content using a simulated identity and an access control mechanism to preserve the privacy of the users. Large data objects, such as images, are stored off-chain with cryptographic hashes anchored on the blockchain. Additionally, an Artificial Intelligence (AI)-assisted content moderation mechanism will be used to minimize the impact of malicious, low-quality reporting content. To demonstrate the feasibility of the proposed framework, A web-based Decentralized App (DApp) is developed as a proof of concept within a controlled experimental environment. Through this project, we aim to deliver a practical and scalable prototype that enhances citizen participation, accountability, and trust in local governance systems.

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Acronyms

AI	-	Artificial Intelligence
API	-	Application Programming Interface
CID	-	Content Identifier
CI/CD	-	Continuous Integration and Continuous Deployment
CNN	-	Convolutional Neural Network
DApp	-	Decentralized Application
FSM	-	Finite State Machine
IPFS	-	InterPlanetary File System
KPI	-	Key Performance Indicator
NGO	-	Non-Governmental Organization
NLP	-	Natural Language Processing
PaaS	-	Platform-as-a-Service
PII	-	Personally Identifiable Information
PoA	-	Proof of Authority
PoS	-	Proof of Stake
PoW	-	Proof of Work
RBAC	-	Role-Based Access Control
RPC	-	Remote Procedure Call
TPS	-	Transactions Per Second
VPS	-	Virtual Private Server
ZKP	-	Zero-Knowledge Proof

Chapter 1

Introduction

Local governance is fundamental to the upkeep of public infrastructure and the overall well-being of the community. To address public problems such as damaged roads, issues in waste management, street light failures, effective communication between citizens and local authorities is critical. Recently, the integration of e-governance systems and crowdsourcing has improve the citizen participation and streamline the reporting and resolution of public issues.
However, most of the current systems rely on centralized architectures. This poses a challenge in dealing with data integrity, transparency, and user privacy. Citizens are reluctant to provide complaints about sensitive topics or criticism due to potential retaliation or the possibility of their personal info being abused. At the same time, the local government is faced with the challenge of verifying the authenticity of anonymous complaints raised by citizens.
Some of the key features of Blockchain technology are immutability, transparency, and decentralization. Those features can be used to address several limitations of conventional reporting systems. It is possible to design reporting platforms that protect user anonymity while ensuring the integrity and accountability of reported information by combining blockchain-based architectures and privacy-preserving mechanisms. This project examines the application of blockchain technology in developing a secure, community-driven reporting framework tailored to local governance contexts.

1.1 Problem Background

In the context of local governance, crowdsourced reporting systems and e-governance platforms are widely adopted. By using those systems, citizens can report issues, provide feedback and participate in decision making. It improves the inclusiveness and the accountability of every citizen. In developing countries like Sri Lanka, these systems have the potential to significantly improve the governance efficiency.

Timely identification and the resolution of public conccerns are the key to governance efficiency.
However, despite their advantages, the conventional e-governance and reporting systems are still highly centralized, making them prone to various threats such as data tampering, illegal access, and single points of failure. In addition, the ability to track behavior down to individual users could serve as a discouraging factor, particularly in reporting delicate matters, in a system that lacks anonymity.
Blockchain-based systems have appeared as a feasible solution for managing shared data in a tamper-proof and transparent manner. The ability of permissioned blockchains to enable controlled participation while maintaining accountability through trusted validators makes them a desirable choice for governance applications. By using smart contracts, reporting workflows and opinion polling can be automated rather than relying on centralized intermediaries.
While blockchain technology offers data integrity and transparency, it still poses challenges related to user privacy, content moderation, and trustworthiness among anonymous participants. These challenges remain open research and implementation problems. Combining blockchain with privacy-preserving identity mechanisms and AI-assisted moderation can help address these limitations and supports the need for a comprehensive framework for secure and reliable community-assisted reporting.

1.2 Problem Statement

The existing frameworks of e-governance and crowdsourced reporting systems for local governance do not provide an integrated mechanism that can guarantee data integrity, user anonymity, and trustworthiness of citizen submitted reports. Centralized architectures are prone to unauthorized access, data tampering, and loss of transparency. At the same time, identity dependent systems discourage citizen engagement due to privacy concerns and fear of retaliation.
Although blockchain has been applied to the context of local government context, many blockchain-based reporting systems still fail to balance anonymity with mechanisms for assessing the reliability and quality of submitted reports. Furthermore, the lack of content moderation and trust evaluation increases the risk of misinformation, and malicious or low quality submissions.
Therefore, there is a clear need for a privacy-preserving, blockchain-based reporting framework that enables anonymous citizen participation while ensuring data immutability, transparent workflows, and reliable evaluation of reported information.
This project aims to address this gap by designing and prototyping a permissioned blockchain-based community-assisted reporting system that incorporates smart contracts, off-chain data storage, and AI-assisted content moderation mechanism

suitable for local governance environments.

1.3 Objectives and Scope

1.3.1 Aim

The aim of this project is to design and prototype a blockchain-based, privacy-preserving, community-assisted reporting system for local governance that empowers citizens to report public issues and provide their opinions in a transparent, accountable, and secure manner.

1.3.2 Objectives

1. Design and implement a permissioned reporting blockchain using a PoA consensus mechanism to ensure the integrity and immutability of citizen-submitted reports.

2. Design and deploy smart contracts to automate public issue reporting workflows, including ticket creation, issue resolution and closure, community voting as feedback on resolved issues, and separate opinion polling mechanisms for public consultation.

3. Develop user-friendly interfaces that enable citizens and authorized government officials to interact with the blockchain for reporting issues and tracking their resolution status.

4. Design and integrate an AI-based content moderation system to detect and filter inappropriate textual and visual content in user-submitted reports.

1.3.3 Scope

1. A prototype of the proposed blockchain will be implemented with a minimum of three full nodes. The PoA consensus mechanism will be used to enable efficient and controlled transaction validation within a permissioned network, while excluding deployment of public, permissionless blockchains or globally distributed ledger infrastructures.

2. User authentication and identity decoupling are managed through a simulated identity service, and the design and implementation of a fully decentralized Self-Sovereign Identity (SSI) blockchain are explicitly excluded from the scope of this project.

3. The integration of AI within the system is limited to content moderation using existing machine learning models, libraries, or external Application Programming Interfaces (Application Programming Interface (API)s), and the development of custom, from-scratch deep learning architectures is not within the scope of this work.

4. Transaction validation authority within the blockchain network is restricted to pre-approved institutional stakeholders, including local government entities and recognized non-governmental organizations, and does not permit anonymous public participation or Proof-of-Work (Proof-of-Work (PoW)) based mining.

5. The project emphasizes functional validation, system evaluation, and performance analysis within a controlled experimental environment, and does not address real-world legal compliance, regulatory adoption, legislative reform, or large-scale integration with existing national government information systems.

6. The user interface development is limited to a responsive, web-based DApp accessible via standard mobile and desktop web browsers, and explicitly excludes the development of native mobile applications (iOS or Android) or cross-platform native frameworks.

1.4 Literature Review

1.4.1 Digital Citizen Reporting and e-Governance Systems

Electronic governance (e-governance) is using the information and communication technologies to provide government services. This includes the support of government to citizens (G2C), government to business (G2B) , government to agencies (G2A) etc. By utilizing the e-governance platforms, everyone can gain access to whole range of public services. Additionally citizens can submit complaints and participate in governance decision making process. [2].
Digital citizen reporting system is one of the many use case of e-governance. "FixMyStreet" is a a such existing platform. It enable citizens to report issues for different issue categories. Sanitation, Traffic and Public Safety are some examples. Previous studies shows that these kind of systems improve governance administrative efficiency, civic participation and also they increase the transparency.[3].
Even though there are many benefits of digital citizen reporting in local governance as described earlier, most of them still use centralized server based architecture. This centralized approach have its own limitations and disadvantages. Susceptibility to data tempering, lack of transparency are the main problems that impact

heavily on public trust. In addition to that, mandatory identity disclosure is discouraging public participation on this kind of reporting systems. Therefore these problems motivate the findings of new decentralized, transparent and immutable architectures that still preserve user privacy with accountability.

1.4.2 Blockchain for Local Governance Applications

Blockchain is a technology that has been popular for its ability to have an immutable, transparent and decentralized ledger for public data. Hence, its a transformative tool for e-governance to address the problems that existing implementations had. The Blockchain allow us to verify the immutability of the records, so its suitable for governance applications where the trust and the accountability is crucial. [4], [5].
Blockchain based land registry system implemented in Cook Country, Illinois is a one real world example of the applicability of blockchain in local governance. They aim to reduce the property fraud by consolidating historical ownership records into a verifiable block in the blockchain. Similar to that Delaware has taken initiative to explore the use of smart contracts that's enable real time asset tracking and automate business fillings. This improves the regulatory compliance and oversights [5].
In addition to administrative efficiency, blockchain has been utilized for citizen centric services, such as secure digital identity management. Estonia has taken a initiative to integrate blockchain across healthcare, taxation and even electronic voting systems. So this can provide a mathematically verifiable audit trail for government data. MyPass project in Austin and the World Food Program's building blocks initiative also use Blockchain technology under the hood to manage encrypted identity and transaction records [4].
Despite the advancements in the blockchain base applications, existing literature highlights the challenges of adoption in governance. Scalability constraints, privacy concerns related to identity data, performance overhead and also the need of the legislative and institutional reforms are some examples. These factors needs to be addressed by carefully selecting blockchain architectures and consensus mechanisms before implementing systems for frequent citizen interactions.

1.4.3 Blockchain-Based Citizen Reporting and Complaint Management Systems

The traditional centralized platforms have data integrity issues, transparency issues and trust issues when it comes to citizen reporting. Past works also indicates that centralized complaint registries often have single point of failures , delayed

responses and public mistrust. Therefore blockchain integrated reporting services can be used to mitigate those issues.
A blockchain based police complaint management system has proposed by Pratheeba [6] contains a modular architecture. It's include security, blockchain, web interface and a hybrid cloud storage. This system ensure immutability of complaint records by cryptographic hash linking. And also it utilize RSA based encryption to restrict the access only to authorized personnel. In Addition to that, Pratheeba used Natural Language Processing (NLP) techniques to automatically categorize complaints in the system based on their priorities.

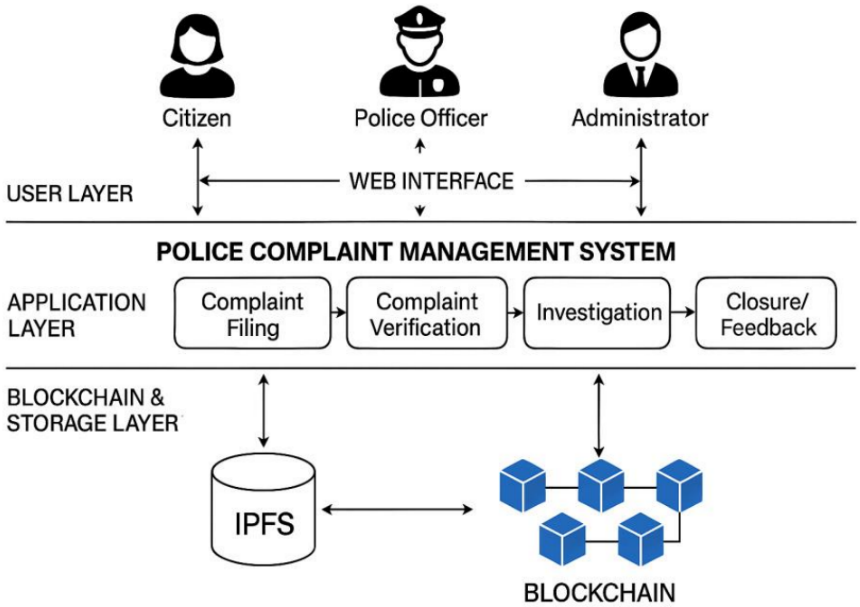

Figure 1.1: System Architecture of the Police Complaint Management System.[1]

A three tier architecture for decentralized complaint management framework has proposed by Manjula [1] is shown in the Figure 1.1. The three layers are web based presentation layer, a business logic layer and finally a blockchain data layer. This system has a role based access control method to involve multiple stakeholders. Citizens, Law enforcement authorities and judicial bodies are some of them. More importantly this system has a immutable storage for multimedia evidences. Intelligent complaint assignment based on workloads, automatic escalation mechanism are some of the key features of the framework. The experimental evaluations shows that a significant reductions in complaint processing time. The successful detection of data tampering also archived by blockchain verification.
In literature, existing blockchain based complaint management systems shows some improvements in transparency and administrative efficiency. But most of them

are domain specific. They primarily target law enforcement or isolated governance functions. In addition to that several platforms still rely on centralized architectures. Only a limited attention is given for implementing a community assisted validation, for privacy preserving reporting mechanisms and efficient handling of multimedia evidences. Therefore these limitations highlight the need for a generalized reporting framework for local governance.

1.4.4 Permissioned Blockchains and PoA Consensus Mechanisms

Based on the participation and access control blockchain network can be classified in to two categories. [7]. They are permissionless architecture (public) and permissioned architecture. Bitcoin and Early Ethereum networks are permissionless blockchains. They allowed unrestricted participation and relyed on consensus mechanism like PoW or Proof of Stake (Proof-of-Stake (PoS))[8]. These architectures and consensus mechanisms emphasized decentralization among participants. But they often suffer from the high latency of the network, limited throughput and unpredictable transaction costs due to congestion. These problems make the traditional permissionless PoW or PoS consensus mechanisms unsuitable for governance and crowdsourced applications[9].
While permissionless blockchains are unsuitable for governance related applications, permissioned blockchains restrict the network participation to authorized entities. This enables more efficient consensus mechanisms and greater control over governance data [7]. This known identities of participants allow permissioned systems to achieve higher performance. And support the regulatory compliances, data privacy and structured governance models as well citemdpi2023unleashing.
Other than PoW and PoS consensus mechanisms, PoA offer significant advantages over permissioned blockchains. PoW requires energy intensive computation. PoS relies on the economic stake and introduce wealth centralization risks. But on the other hand PoA depends on the identity and the reputation of pre approved validators[10], [11]. Therefore this PoA consensus mechanism enables high transaction throughput and low latency. And also it mitigate the issues such as the "nothing at stake" problem by off chain legal and reputational accountability [12], [13].
The government systems should have accountability, performance predictability and data confidentiality. The PoA is well suited consensus mechanism for that [14]. By using PoA we can have the ability to identify and regulate validator entities to align blockchain operations. It's because we have the existing legal frameworks that ensure institutional responsibility. Hence, can avoid malicious behavior [15]. In Addition to that PoA avoids the indeterministic transaction fees and support controlled validation environments. This make them viable for handling sensitive

governmental data under regulatory constraints[11],[16]. As a result, PoA emerges as a practical consensus mechanism for permissioned blockchain-based reporting systems in local governance contexts.

1.4.5 Smart Contracts for Public Issue Management and Opinion Polling

In governance systems, to improve the transparency, efficiency and citizen participation, smart contracts have been explored as the good mechanism. Smart contracts can automate the governance rules and incentives by immutable code. It can support models like civic intelligence governance (CIG). In CIG, citizen engagement is encouraged through on chain participation and token based incentives[17]. Literature shows that approaches like CIG can improve participation rates and knowledge sharing while reducing relances on centralized intermediaries. In the context of multi organizational governance smart contracts also been used within dual blockchain architectures. This method separates transaction data from contract logic. It balance the transparency and privacy in Government to Government and Business to Business processes[18].
Smart contracts also support business processes in governance by embedding policy rules directly into executable logic. Bid submission, evaluation and contract execution are some of the use cases in the public procurement and tendering processes that smart contract hold its value. It also reduce the oppertunities for corruption and increase the accountability[18], [19], [20]. Similar benefits are observed in the blockchain based land registry systems. Smart contracts have enable immutable property records automated ownership transfers and enforcement of land use policies. This has reduced fraud and administrative inefficiencies[21], [22].
In the context of voting and opinion polling, smart contracts offer enhanced security, transparency, immutability and verifiability inherited from blockchain[23], [24]. There are advance mechanisms such as Quadratic Voting and also blockchain enabled participatory budgeting. Those mechanisms further extend citizen involvement in decision making. It have expressive preference representation and transparent allocation of public resources[25], [26]. In addition to these advantages, there exists some challenges related to scalability, privacy and system complexity when deploying voting and polling systems to large and diverse populations[23], [27].

1.4.6 User Interfaces for Blockchain-Based e-Governance Applications

User interfaces play a critical role in determining the accessibility and adoption of blockchain based e-governance systems. In the traditional applications the user literacy is less important but in this conventional blockchain applications the user literacy is highly considered because of the blockchain concepts that are different from the traditional apps and it does make the users hard to understand and adopt. The literature consistently identifies usability and user experience challenges as major barriers to the adoption of decentralized applications for the general public. Particularly where security requirements conflict with established usability principles [28], [29]. So the design of intuitive, inclusive, and responsive web interfaces is essential for enabling effective interaction between citizens, government officials, and the underlying blockchain infrastructure.

Usability Challenges in Decentralized Governance Systems

Decentralized applications are fundamentally different from traditional web systems due to its architectural characteristics. Blockchain based systems have asynchronous transactions, immutable records, and cryptographic identity management. They can create friction for users who are used to centralized platforms. Studies highlight a significant usability gap in Web3 applications because of the concepts like transaction finality, gas fees, and private key ownership which are difficult for non technical users to understand and manage [28], [29].

Identity Management and Cognitive Load

One of the most critical usability barriers is wallet based identity management. The reason is user identity is tied to cryptographic key pairs instead of institution managed accounts In decentralized systems. Research shows that the responsibility of safeguarding private keys and the unrecoverble key loss red ce the user participation. Specially in general public where inclusivity is essential [29], [30], [31].

The literature further shows that concepts such as self-sovereign identity and identity management remain poorly understood by most users. Studies report unsafe practices and misunderstanding of recovery mechanisms when users are presented with seed phrases or key-based authentication [32], [33]. These findings motivate the need for interface-level abstractions that preserve blockchain security while reducing cognitive load for citizens and officials.

Transaction Latency and User Feedback

Blockchain transactions are inherently asynchronous with confirmation times dependent on network conditions and consensus protocols. This latency can lead to confusion among the users because of the system status problems and users can determine those as system failures [29]. The literature recommends to use an optimistic user interface pattern, where user actions are acknowledged immediately while blockchain confirmation occurs in the background.
As advanced design solutions, literature shows progressive status disclosure, distinguishing between off-chain receipt of a report and on-chain verification which allows users to track the lifecycle of an issue without being exposed to unnecessary low level blockchain details [34], [35]. Mechanisms like that are particularly relevant for issue reporting systems that require transparency and trust.

Transaction Costs and Abstraction of Blockchain Complexity

A significant usability challenge coming from the blockchain is transaction cost that associated with public blockchains. And also the concepts of gas fees and fluctuating gas costs are very unfamiliar in public services context. Most of the time users expect free access to the reporting platforms[28]. The literature strongly suggest that the use of gas fee abstraction techniques. Meta Transactions is one of the example. In Meta, backend relayer services submit transactions on behalf of the user. Its achieved by preserving cryptographic intent verification[29]. Therefore this approach enable citizen to interact with blockchain without technical or economical complexities.

Comparison with Centralized Reporting Platforms

FixMyStreet and SeeClickFIx are existing centralized platforms. Eventhough they are centralized, they provide a valuable benchmarks for crowdsource citizen reporting interfaces. Inituitive map based reporting, simple authentication mechanisms and clear feedback loops are the key characteristics that attributed to platform success[36], [37].
With addition to that SeeClickFix further uses a gamification mechanism to encourage the users to participate in the system. In contrast, Even though the proposed application uses blockchain for reporting platform, it must retain the usability and the expectations of the users. The literature highlights the need of interface design such that gurantees the data integrity and auditability using familiar ui elements rather than cryptographic representations[38], [35], [39].

Accessibility and Inclusivity Considerations

Inclusivity is a fundamental requirement for e-governance systems. According to [40], it can be seen that poorly designed blockchain interfaces may digitally divide users. Also it may favor crypto-literate users excluding vulnerable populations. To reduce this issue, literature suggests mobile first design approaches. Furthermore, lightweight client architectures, effiecient bandwidth synchronization can used to support access in resource limited environments. [41].

1.4.7 AI-Based Content Moderation in Civic Reporting

Since blockchain is inheritably immutable its introduce a unique challenge for content moderation in a civic reporting system. In a centralized platforms, administrators can remove or edit inappropriate content. But in a decentralized blockchain based system data cannot be altered once confirmed [42]. While immutability strengthens transparency and accountability, it also raises the risk of permanently storing harmful, illegal, or misleading content. Therefore the literature focuses the need for pre moderation strategies to prevent unsuitable content from being recorded on chain or to limit its visibility after publication [34].

1.4.8 Need for Automated Content Moderation

Civic reporting platforms are susceptible to several forms of misuse that can compromise system reliability and public trust. Automated spam submissions, the submission of malicious or legally sensitive content, and deliberate dissemination of false or misleading reports are some of the common issues [43], [44], [45]. Manual moderation is impractical for above situations. Since the reporting system is large latency, human bias, and operational costs can occur. Furthermore it reintroduces the centralized control structure that blockchain system is aim to minimize.
Therefore the literature supports AI assisted moderation as a scalable and efficient approach for filtering content before its permanently recorded. It maintain the acceptable levels of decentralization [46], [47].

1.4.9 Architectural Approaches for AI Blockchain Integration

Due to the computational and storage limitations of blockchain platforms, AI-based moderation cannot be performed directly on chain. Off chain processing combined with on chain verification identified by existing researches as most practical approach[47], [48].

In this approach, reported content is first analyzed by AI models hosted off chain infrastructure or decentralized compute networks. The outcome of this analysis is a validation decision or a trust score. This outcome then submitted to the blockchain as a cryptographic proof or oracle response. Next the smart contracts verify those results before accepting the report. It will ensure that only the content meeting predefined policies are recorded on chain. BlockCITE is a example framework demonstrate how this separation preserves blockchain integrity while avoiding excessive on chain computation [49].

1.4.10 AI Oracles and Decentralized Oracle Networks

Oracles are trusted intermediaries that connect on chain and off chain. In here the AI oracles provides the computational results to smart contracts. Instead of simply retrieving external data to the blockchain, AI oracle perform content classification and verification tasks. Then it returns the structured results to the blockchain [50], [51].
There can be a risk of oracle bias or failure. To minimize the risk a decentralized oracle networks aggregate outputs from multiple independent AI nodes and determine consensus outcomes. This model reduces reliance on a single AI system. Therefore it improves te robustness by requiring agreement among multiple validators[51], [52].

Techniques for Content Analysis

AI-based moderation techniques vary based on the type of submitted content. NLP models are widely used to detect spam, hate speech and misinformation for text based reports. Studies report high accuracy when combining deep learning models with semantic embeddings. It makes them suitable for filtering large volumes of text submissions prior to blockchain recording [43].
For image based reports computer vision models are used to identify inappropriate contents. However recent advances in generative AI increase the risk of fabricated images. This literature proposes a hybrid verification pipeline that combine cryptographic provenance mechanisms with AI-based image analysis. This ensue that the submitted media is both authentic and contextually valid [53].

Hybrid Human AI Moderation Models

AI enable the scalable moderation. But it may struggle with contextual interpretation and edge cases. Because of that many studies advocate hybrid moderation models. In those models AI performs initial filtering and prioritization, while human reviewers handle disputed or ambiguous cases [46], [54].

Decentralized community based moderation further complement this approach. The models using community annotation systems, allow users or designated validators to challenge AI decisions during predefined review periods. In Addition to that dispute resolution handled by transparent governance mechanisms [34], [55]. Therefore these strategies reduce the reliance on central authority while maintain the accountability.

Ethical Considerations and Content Permanence

Ethical concerns are crucial when implementing systems by integrating AI driven moderation systems. The algorithmic bias and the accountability are the highlights in the literature. Biased training data can lead to unfair rejection of legitimate reports[56]. To minimize this risk explainable AI mechanisms are recommended. This enable users to understand the moderation decision and resubmit corrected reports [57].
Finally, the issue of "toxic immutability" remains a critical challenge for blockch ain-based systems. To resolve this issue its suggested that to use a off chain storage systems like IPFS. If the content is later identified as illegal or harmful, it can removed from the off chain storage. But can preserve the immutable audit trail by recording it on the blockchain[53]. The transparency, legal compliance and ethical responsibility can be balanced by this approach.

1.5 Report Outline

This report is organized into six main chapters. Each chapter focuses on a different part of the project, from the problem background to the current implementation progress and future work.
Chapter 1 introduces the research problem and explains why a privacy-preserving reporting system is needed for local governance. It presents the problem background, problem statement, objectives, scope, and literature review related to citizen reporting systems, blockchain-based governance, AI moderation, and decentralized applications.
Chapter 2 provides the technical background required to understand the proposed system. It explains the main technologies used in the project, including blockchain, permissioned blockchain with Proof-of-Authority consensus, smart contracts, DApps, IPFS, identity privacy concepts, and AI-based content moderation.
Chapter 3 describes the methodology and implementation of the proposed framework. It explains the research approach, system requirements, overall architecture, workflow, project timeline, and expected outcomes. It also describes the implementation progress of the main components, including the private blockchain, smart

contracts, identity service, backend relayer, IPFS integration, AI moderation service, and web-based DApp.
Chapter 4 presents the testing, experimentation, and validation approach. It describes the experimental setup, testing methodology, subsystem testing, integration testing, and preliminary validation of the implemented components. It also discusses the current results and compares the proposed approach with existing reporting systems.
Chapter 5 discusses the main findings of the project. It highlights the important contributions of the proposed methodology and implementation, evaluates the progress against the project objectives, and explains the limitations of the current prototype.
Chapter 6 concludes the report by summarizing the overall progress and significance of the project. It also presents the future work required to complete and improve the system, including full integration, image moderation, government-side workflows, stronger identity mechanisms, and further testing.

Chapter 2

Background

2.1 Blockchain Technology

Blockchain is a distributed, digital, append-only ledger technology. It enables data to be stored in immutable and transparent manner across multiple nodes in a network. Unlike centralized systems, blockchain systems maintain a syncronized copy of the ledger among all authorized participants. This approach reduces the dependency on a single intermediary controlling authority. Each block in the blockchain contains a set of validated transactions, a timestamp, and a cryptographic hash of the previous block. This hash-chaining mechanism ensures that any alteration to previous recorded data becomes immediately detectable.
Immutability is a key characteristic in blockchain technology. It is computationally impractical to alter or delete a transaction after it is confirmed and appened to the blockchain. The immutable property is significantly important in governance-related systems, because integrity and transparency are a must.
Cryptography plays a significant role in blockchain systems. To generate unique digital fingerprints of data, hash functions are used. To verify ownership and authenticity of transactions, digital signatures are used. By using above cryptographic methods, blockchain systems gurantee integirty, authenticity and non-repudiation of data.

2.2 Permissioned Blockchain and PoA

Bitcoin and Ethereum are permissionless blockchains, where anyone can participate as a node and validate transactions. In contrast, permissioned blockchains restrict participation to a predefined set of entities. These entities, often referred to as validator nodes. They are responsible for validating transactions and maintaining the integrity of the blockchain.

While permissionless blockchains rely on consensus mechanisms like PoW or PoS, the PoA consensus mechanism is specially designed for permissioned blockchains. In here instead of relying on computationally expensive mining or staking process the validator nodes in the permissioned blockchain are validating blocks. Therefore this approach is a reputation and indetity-based mechanism, where the authority of the validators is based on their identity and reputation within the network. This allows for faster transaction processing and lower energy consumption compared to traditional consensus mechanisms, making it suitable for enterprise applications and private networks.
In the context of local governance perspective, such system requires predictable performance, immutability, transparency and accountability. PoA consensus running in a permissioned blockchain is well suited for this requirements because validators identities are known and they controlled by trusted entities. With addition to that, PoA eliminates the need for economic incentives. This make it more efficient and practical for local governance applications.

2.3 Smart Contracts

Smart contracts are first introduced by Ethereum blockchain. They are self-executing contracts with the terms of the agreement directly written into code. In a blockchain based governance systems smart contract can be used to eliminate centralized administrative control. It can automate workflows that would traditionally require human intervention. Therefore this reduce the reliance on intermediaries, hence improve the transparancy and trusthworthiness.
The proposed system uses Solidity based smart contracts deployed on a private permissioned PoA blockchain.

2.4 Decentralized Applications (DApps)

DApps are software applications that operate on a decentralized network, such as a blockchain, rather than being hosted on centralized servers. Unlike traditional web applications that rely on centralized databases and backend infrastructure, DApps utilize smart contracts to handle backend logic and state management. This shift from traditional Web2 to Web3 architecture ensures that application data is immutable, transparent, and resistant to single points of failure or centralized control.
In a typical DApp architecture, the frontend provides the user interface, while the core logic is distributed across the blockchain network. To interact with the blockchain, the frontend utilizes Web3 provider libraries, such as Web3.js or Ethers.js. These libraries enable the application to communicate with blockchain

nodes via Remote Procedure Calls (Remote Procedure Call (RPC)). Through this connection, the frontend can query the blockchain to read the current state of smart contracts or construct new transactions. When a user wishes to modify the blockchain state, they must cryptographically sign the transaction using their digital wallet's private key, ensuring authenticity and authorization.
A significant challenge in traditional DApp adoption is the requirement for users to pay transaction fees, commonly known as "gas," using the native cryptocurrency of the network. This creates a barrier to entry for regular users who may not possess or understand cryptocurrency. To address this usability issue, modern DApps employ meta-transactions and relayers. A meta-transaction allows a user to cryptographically sign their intent to execute a transaction off-chain without broadcasting it themselves. This signed message is then sent to an intermediary service known as a relayer. The relayer, which holds a balance of the network's cryptocurrency, wraps the user's signed message in a standard blockchain transaction, submits it to the network on behalf of the user, and pays the associated gas fees. This approach provides a seamless, "gasless" user experience while maintaining the security and non-repudiation guarantees of cryptographic signatures, making blockchain applications more accessible to the general public.

2.5 InterPlanetary File System (IPFS)

InterPlanetary File System (IPFS) is a peer-to-peer network protocol designed for storing and sharing files in a decentralized manner. Unlike the traditional web, where files are identified and accessed through server-based addresses such as URLs, IPFS locates files based on their content. This content-addressed approach improves resilience, reduces dependence on centralized servers, and provides an efficient off-chain storage mechanism for blockchain-based applications. A fundamental concept in IPFS is the Content Identifier (Content Identifier (CID)), which is a cryptographic hash generated from the content of a file. If the file content changes, the CID also changes, ensuring strong integrity guarantees because users can verify that the retrieved content exactly matches the expected CID. Since identical content generates the same CID, IPFS also supports deduplication and immutable references. In the IPFS network, data can be hosted by any node that chooses to store or pin the content, allowing files to be retrieved from multiple peers using their CID. This decentralized storage model increases availability, improves fault tolerance, and reduces reliance on a single server, although long-term persistence depends on nodes continuing to pin the data. In blockchain-based systems, IPFS is commonly used as an off-chain storage layer because storing large files directly on-chain is costly and inefficient. Instead, applications store only CIDs and essential metadata on the blockchain, while the actual content remains in IPFS. Users can then retrieve the file through its CID and verify its authenticity using cryptographic hashing. Overall, IPFS provides a scalable and cost-effective decentralized storage solution that combines content integrity, distributed hosting, and efficient off-chain data management.

2.6 Zero-Knowledge and Identity Concepts

Zero-knowledge proofs allow a user to prove that they possess sensitive information without actually revaling the information itself.
The identity decoupling is critical requirment because citizens may hesitate to report sensitive problems they have if their identity is linked to the submitted report. Therefore we decouple identity of the citizen by using public-private key pair associated with the citizen. By leveraging this approach the blockchain ledger will record only pseudonymous identifiers, authorization tickets and cryptographic proofs. This ensure no sensitive personal information is ever oublically exposed.

2.7 AI-based Content Moderation

Citizen reporting platforms are vulnerable to spam Submissions, abusive content, misinformation, and malicious reports. Due to immutability of blockchain records, permanently storing harmful content presents significant ethical and operational concerns. Therefore, content moderation techniques are required before reports are recorded on-chain.
AI techniques are widely used for automated content moderation. NLP techniques are used in text moderation tasks such as:toxicity detection, hate speech detection, spam filtering, and sentiment Analysis. For image moderation, computer vision and deep learning approaches are mostly used. Convolutional Neural Networks (Convolutional Neural Network (CNN)s) and transformer-based vision models can classify images based on categories like: explicit content, graphic violence, hate symbols, and unrelated media.

Chapter 3

Methodology and Implementation

3.1 Methodology

3.1.1 Research Methodology

This project follows a design-oriented and prototype-based research methodology. The main objective is to design, implement, and evaluate a blockchain-based, community-assisted, privacy-preserving reporting framework for local governance. The system is developed as a functional prototype within a controlled experimental environment rather than as a large-scale production deployment.

The research approach is based on identifying the limitations of existing centralized citizen reporting platforms and designing a decentralized alternative that improves transparency, integrity, anonymity, and trustworthiness. The project combines several technologies, including a private permissioned blockchain, smart contracts, a simulated identity verification mechanism, off-chain storage, AI-assisted content moderation, and a web-based decentralized application.

The system is developed using an incremental and modular development approach. Each major subsystem is designed and implemented independently before being integrated into the complete reporting workflow. This reduces complexity and allows each team member to focus on a specific technical component while maintaining compatibility with the overall architecture.

The major system components are:

1. Web-based DApp
2. ZKP GovID simulator
3. Backend relayer
4. AI moderation oracle service

5. IPFS off-chain storage
6. Smart contracts
7. Private permissioned PoA blockchain

Each subsystem is first implemented and tested individually. After subsystem-level validation, the components are connected through defined APIs, blockchain interfaces, and data formats. The final integration focuses on the end-to-end report submission scenario, where a citizen submits a report through the DApp, the report is validated and moderated, the content is stored off-chain, and the final metadata is recorded on-chain.
This approach follows the phased methodology described in the project proposal, where requirement analysis, system design, blockchain implementation, smart contract development, identity integration, IPFS integration, AI moderation, DApp development, testing, and evaluation are carried out incrementally.

3.1.2 System Requirements Analysis

Functional Requirements

The functional requirements define the main services that the system must provide.

1. Citizen authentication: A citizen should be able to authenticate using the simulated GovID identity service.
2. Ticket issuance: After successful verification, the identity service should issue a citizen seed and a batch of one-time ticket IDs.
3. Report creation: A citizen should be able to create a report containing text, category, location, timestamp, and optional media.
4. Payload signing: The DApp should sign the report payload using the citizen's private key before sending it to the relayer.
5. Relayer verification: The backend relayer should verify the ticket, citizen signature, payload hash, and file constraints.
6. AI moderation: The report content should be analyzed by AI moderation services before blockchain submission.
7. Off-chain storage: Accepted report content and media files should be uploaded to IPFS.

8. Blockchain submission: The relayer should submit accepted report metadata to the reporting smart contract.

9. Report retrieval: The DApp should allow citizens and authorized users to read submitted reports and their statuses from the blockchain.

10. Government action workflow: The system should support future integration of government-side actions such as issue review, resolution updates, and closure.

Non-functional Requirements

The system also has several non-functional requirements.

1. Security: The system should use hashing, digital signatures, ticket validation, role-based access control, and secure API communication.

2. Privacy: Citizen identities should be decoupled from on-chain report records.

3. Integrity: Report payloads, IPFS content, oracle decisions, and blockchain records should be tamper-evident.

4. Availability: The deployed services should remain accessible during testing and demonstration.

5. Performance: The system should process reports within reasonable time under prototype workloads.

6. Maintainability: The system should be modular and containerized to simplify deployment and debugging.

7. Auditability: Important workflow decisions should be logged and anchored through hashes or blockchain events.

3.1.3 Proposed System Architecture

Overall System Architecture

The proposed system uses a hybrid decentralized architecture that combines on-chain and off-chain components. The blockchain is responsible for storing immutable metadata, workflow states, and references to report content. Computationally intensive and large-data operations are handled off-chain through the relayer, AI moderation service, and IPFS. The overall architecture of the system is shown in the Figure 3.1

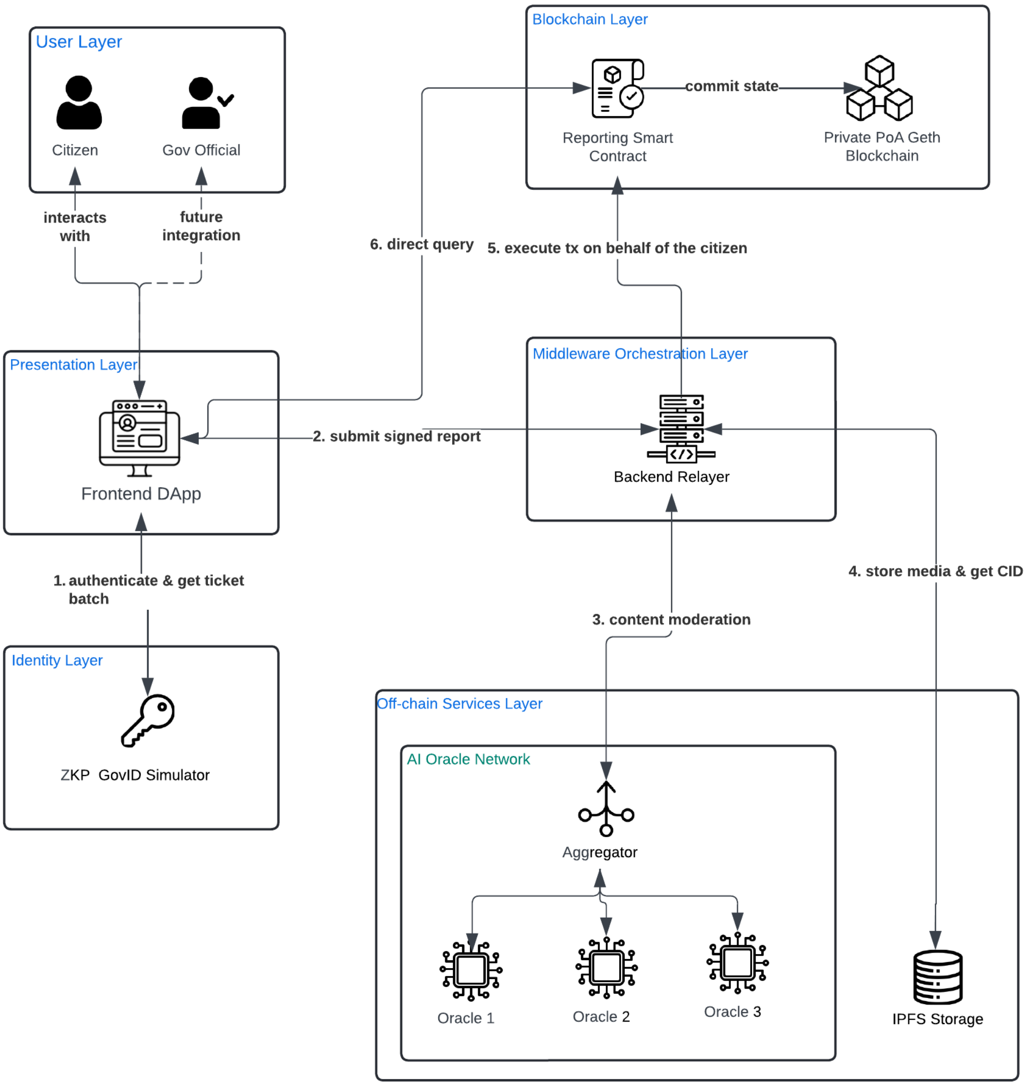

Figure 3.1: High Level Architecture Diagram.

The DApp acts as the citizen-facing interface. The ZKP GovID simulator handles identity verification and ticket issuance. The backend relayer verifies signed report submissions and coordinates communication between AI moderation, IPFS, and smart contracts. The AI moderation service evaluates report content before storage and blockchain submission. IPFS stores large report data and media evidence, while the smart contracts manage the report lifecycle on the private PoA blockchain.
This architecture supports the main project goal of preserving citizen anonymity while maintaining report integrity and transparency. The use of a relayer also improves usability because citizens do not need to directly manage gas payments or manually submit blockchain transactions.

Layered Architecture

The system can be divided into six major layers.

1. User Layer: This layer consists of the citizens and government officials.

2. Presentation layer: This layer consists of the DApp used by citizens and government officials. It provides interfaces for login, report submission, report viewing, and future government-side actions.

3. Identity and authentication layer: This layer consists of the ZKP GovID simulator. It verifies citizens and issues citizen seeds and ticket batches. This separates real identity verification from on-chain activity.

4. Middleware orchestration layer: The backend relayer acts as the coordination layer. It receives signed report payloads, performs validations, calls AI moderation services, uploads accepted content to IPFS, and submits blockchain transactions.

5. Off-chain services layer: This layer includes the AI moderation service and IPFS storage. These components handle tasks that are unsuitable for direct blockchain execution.

6. Blockchain layer: This layer consists of the smart contracts and the private PoA blockchain. It stores immutable metadata, report states, content hashes, IPFS CIDs, and workflow events.

3.1.4 Proposed Workflow

The proposed architecture operationalizes the decentralized reporting framework through a series of distinct, cryptographically secure workflows. These workflows ensure that civic participation remains entirely anonymous to the public and the smart contract, while simultaneously guaranteeing that only verified citizens can submit legitimate data.

Citizen Authentication and Identity Workflow

The authentication workflow bridges real-world civic identity with anonymous on-chain participation without exposing Personally Identifiable Information (Personally Identifiable Information (PII)) to the blockchain network. This is achieved through the integration of the ZKP-GovID Simulator, which acts as the centralized trust anchor for identity verification. The workflow is shown in Figure 3.2.
The step-by-step authentication process executes as follows:

1. The citizen accesses the decentralized application frontend and inputs their official national identification number and password.
2. The frontend transmits these credentials over an encrypted channel to the ZKP-GovID Simulator for verification against the mock government registry.
3. Upon successful credential validation, the simulator generates a predefined batch of randomized, single-use ticket identifiers. These identifiers contain no personal data and serve purely as cryptographic tokens.
4. The simulator utilizes the official Government Authority private key to cryptographically sign each randomized ticket identifier.
5. The generated batch of signed tickets, along with a citizen seed value, is returned to the decentralized application and stored securely in the local browser environment. These tickets function as anonymous, verifiable hall-passes for future platform interactions.

Overall System Lifecycle Workflow

The overall system workflow synthesizes the individual operational pipelines into a singular, comprehensive lifecycle of a civic issue. It maps the entire trajectory of a report from its initial conception by a citizen to its final, verified resolution by a government authority. This macro-level view highlights the continuous interplay between off-chain middleware, decentralized storage, the artificial intelligence oracle, and the on-chain finite state machine. The complete lifecycle is illustrated in Figure 3.3.
The progression of a civic report executes sequentially as follows:

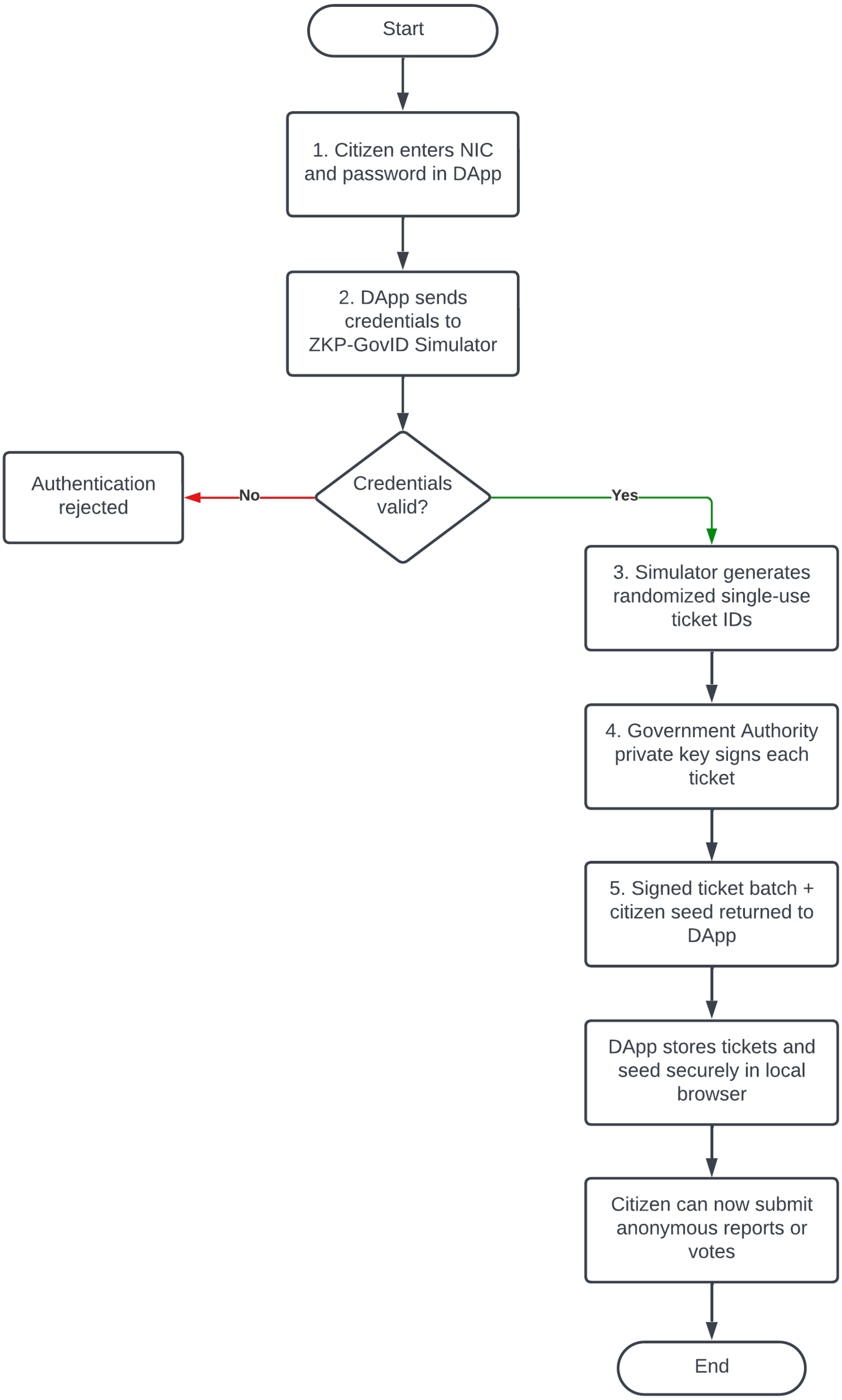

Figure 3.2: Citizen Authentication and Identity Workflow

1. The lifecycle initiates with the citizen authenticating their real-world identity through the centralized simulator to securely acquire a batch of anonymous cryptographic tickets.

2. Utilizing an acquired ticket, the citizen constructs and submits a civic issue report accompanied by a cryptographic signature.

3. The backend relayer intercepts the payload, routing it through the AI moderation ensemble to filter malicious content, and subsequently uploads the approved media to the decentralized storage network.

4. The relayer formulates the blockchain transaction and anchors the lightweight metadata to the smart contract, placing the report into the initial pending validation state.

5. The local community evaluates the pending report. Reaching the rejection threshold terminates the issue, while reaching the validation threshold transitions the report to an open state, formally notifying the government authority.

6. The assigned authority acknowledges the validated issue and triggers a state transition to indicate that active, physical repairs are currently in progress.

7. Upon concluding the physical intervention, the authority updates the smart contract to mark the issue as solved, which temporarily shifts the report into a pending verification state.

8. The community conducts a final, physical review of the repaired site. An acceptance consensus transitions the report to the terminal closed state, whereas a rejection consensus reopens the issue, mandating the authority to rectify the inadequate repair.

Report Submission Workflow

The report submission workflow represents the core operational pipeline of the framework. It delegates heavy computational tasks, such as content moderation and file storage, to off-chain middleware, ensuring the blockchain remains a lightweight and gas-efficient state machine.
When a citizen initiates a civic issue report, the system executes the following deterministic pipeline:

1. The citizen drafts a description of the civic issue and optionally attaches multimedia evidence via the frontend interface.

2. The frontend application selects one unused signed ticket from the local batch. It generates a cryptographic hash combining the issue description and the ticket identifier, and subsequently signs this hash using the citizen's local wallet private key.

3. The application transmits the complete payload, including the raw text, media files, government-signed ticket, and the citizen's signature, to the backend relayer via a secure HTTP request.

4. The backend relayer performs a strict dual-layer cryptographic verification. First, it fetches the Government Authority public key to verify the authenticity of the ticket. Second, it verifies the citizen's signature against the payload hash to detect any tampering during transit.

5. If cryptographic verification succeeds, the relayer routes the submitted text and media to the AI moderation ensemble. The centralized aggregator node processes the inputs through its algorithmic nodes to determine if the content is appropriate or malicious spam.

6. Upon receiving an approval verdict from the AI oracle, the backend relayer offloads the multimedia files to IPFS. The IPFS network returns an immutable CID representing the stored files.

7. The relayer formulates a blockchain transaction containing only the lightweight IPFS CID and the ticket identifier. To ensure a gasless experience for the citizen, the relayer signs this transaction with its own wallet and pays the necessary network fees.

8. The Reporting smart contract receives the transaction. It immediately checks the provided ticket identifier against its internal nullifier mapping to prevent replay attacks and duplicate submissions.

9. Once the nullifier is marked as consumed, the smart contract successfully anchors the report to the ledger, initializing it in the Pending Validation state to await community consensus.

Community Consensus and Validation Workflow

This workflow outlines the mechanism by which the decentralized community filters out spam and validates the legitimacy of submitted civic issues. It ensures that only genuine reports escalate to the government authorities, thereby optimizing resource allocation. The workflow is shown in Figure 3.4.
The validation process executes through the following steps:

1. A newly submitted report enters the Pending Validation state on the smart contract and becomes visible on the community feed within the decentralized application.

2. A registered citizen physically observes the issue in the real world and decides to corroborate the report.

3. The citizen utilizes the application to cast an upvote. To ensure Sybil resistance, the application binds a new, unused Zero-Knowledge ticket from the citizen's local batch to this specific vote and signs the payload.

4. The backend relayer intercepts the voting payload, cryptographically verifies the ticket and signature, and submits the vote to the smart contract.

5. The smart contract verifies that the ticket nullifier has not been used for this specific voting phase of this specific report.

6. The contract increments the vote tally. Once the tally surpasses the predefined validation threshold, the smart contract automatically executes a state transition, moving the report from Pending Validation to the Open state, making it actionable for authorities.

Authority Resolution and Action Workflow

This workflow defines how authorized government or non-governmental organization nodes interact with the validated civic issues on the blockchain. It establishes the transparency of the physical repair process.
The authority interaction proceeds as follows:

1. The government authority monitors the blockchain state via a dedicated dashboard, filtering for reports that have successfully reached the Open state.

2. Upon dispatching a physical repair crew to the location, the authority submits a transaction to the smart contract invoking the state change function.

3. The smart contract validates the transaction signature against its Role-Based Access Control (RBAC) list to ensure the sender possesses the required authority permissions.

4. Upon permission validation, the contract updates the report state from Open to In Progress, providing real-time transparency to the observing community.

5. Once the physical repair is concluded, the authority submits a subsequent transaction to mark the issue as solved.

6. The smart contract processes this transaction and transitions the report into the Pending Verification state, temporarily halting the authority's involvement until the community responds.

Post-Resolution Verification and Appeal Workflow

The final workflow enforces the system's core accountability mechanism. It guarantees that government authorities cannot unilaterally close a report without the community physically confirming that the job was completed satisfactorily.
The verification process executes as follows:

1. Citizens in the vicinity of the purportedly resolved issue inspect the physical location to confirm the repair quality.

2. Citizens utilize the decentralized application to cast a verification vote (accepting or rejecting the repair), utilizing the same secure, ticket-based nullifier mechanism to ensure one-citizen-one-vote integrity.

3. The backend relayer processes and forwards these verification votes to the smart contract.

4. If the positive verification votes reach the required threshold, the smart contract transitions the report to the terminal Closed state, permanently archiving the successful resolution.

5. Conversely, if the negative votes reach the rejection threshold (indicating a poor or non-existent repair), the smart contract transitions the report to the Reopened state.

6. A transition to the Reopened state forces the report back onto the authority's active dashboard, mandating further physical intervention and a repetition of the resolution workflow.

3.1.5 Project Plan and Timeline

Project Plan

The project plan is organized according to the main implementation phases of the system. Each phase focuses on a specific part of the proposed framework and contributes to the final integrated prototype. Table 3.1 summarizes the major phases, key activities, expected outputs, and current progress status.

Table 3.1: Project plan and current progress

Phase	**Main Activities**	**Expected Output**	**Current Status**
Background Study and Literature Review	Study existing e-governance systems, blockchain-based reporting platforms, PoA consensus, IPFS, AI moderation, and DApp usability.	Identification of research gap and technical foundation.	Completed
Requirement Analysis	Define functional and non-functional requirements for citizens, government officials, relayer, blockchain, storage, and moderation components.	Requirement specification and use-case understanding.	Completed
System Design	Design overall architecture, component interactions, report submission workflow, identity model, and deployment architecture.	High-level system design and workflow diagrams.	Completed
Permissioned Blockchain Setup	Configure private Geth blockchain, PoA consensus, validator nodes, and RPC access.	Working private permissioned blockchain network.	Completed
Smart Contract Development	Implement report creation, report state management, access control, ticket validation, and event emission.	Reporting smart contract for managing issue lifecycle.	In Progress
Identity and Access Control	Implement simulated GovID verification, citizen seed generation, ticket batch generation, and pseudonymous key generation.	Privacy-preserving identity and access control mechanism.	Partially Completed
Backend Relayer Development	Implement payload validation, citizen signature verification, ticket validation, oracle communication, IPFS upload, and smart contract transaction submission.	Middleware service connecting DApp, oracle, IPFS, and blockchain.	In Progress

Continued on next page

Table 3.1 continued

Phase	**Main Activities**	**Expected Output**	**Current Status**
AI Oracle Moderation Service	Implement three oracle workers, aggregator, voting logic, decision hashing, and containerized VPS deployment.	Deployed AI moderation service for text-based report validation.	Completed for Text Moderation
IPFS Off-chain Storage Integration	Upload report content and media to IPFS, retrieve CIDs, and verify file integrity using hashes.	Off-chain storage mechanism for report evidence.	In Progress
Web-based DApp Development	Develop citizen login, report submission, report viewing, category selection, and future government-side interfaces.	User-facing web application.	In Progress
Subsystem Testing	Test DApp, smart contracts, AI oracle service, GovID simulator, IPFS, and relayer independently.	Verified subsystem behavior.	In Progress
System Integration and Evaluation	Integrate all components and test the full report submission workflow from DApp to blockchain.	Functional end-to-end prototype and evaluation results.	Pending
Final Demonstration	Prepare final system demo, report, and presentation.	Demonstrable prototype and final documentation.	Pending

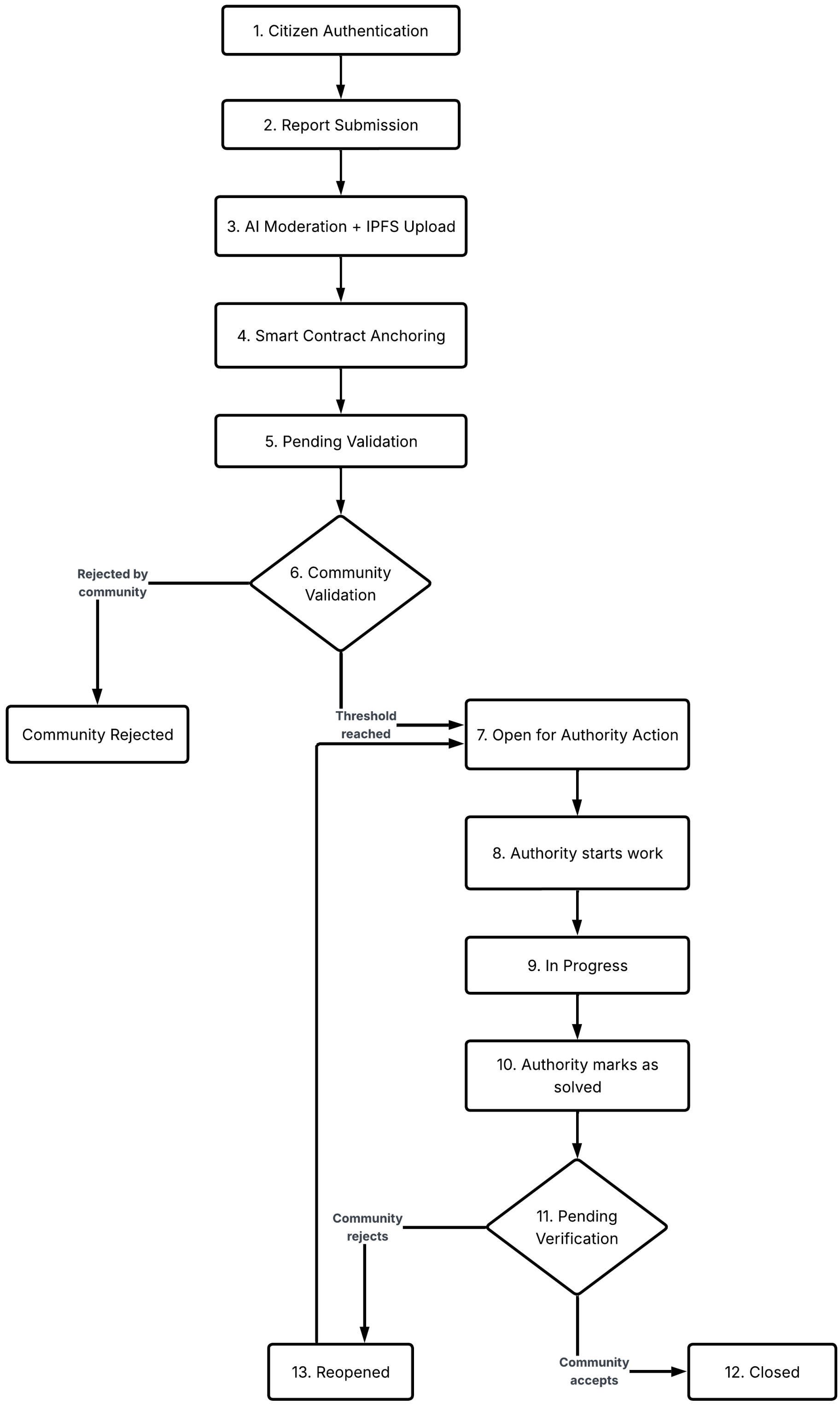

Figure 3.3: Overall lifecycle workflow of a civic issue within the decentralized framework.

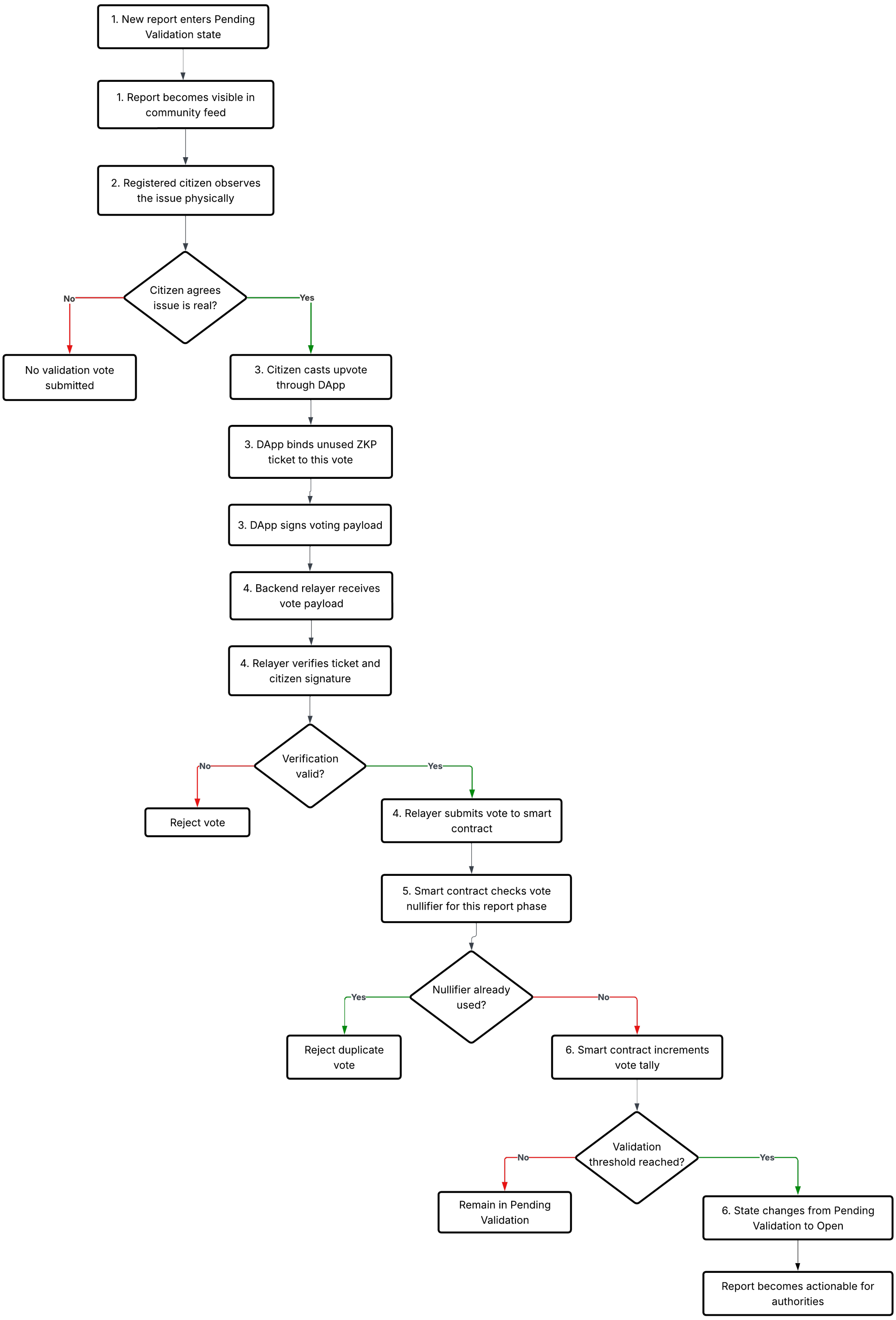

Figure 3.4: Community Validation Workflow

Project Timeline

The project is planned to be completed over two academic semesters using an incremental development approach. The work is divided into several phases, starting from background study and requirement analysis, followed by system design, subsystem implementation, integration, testing, and final evaluation.
The timeline shown in Figure 3.5 was prepared to guide the development process and to allocate sufficient time for each major component of the system. Since this is a progress report, some implementation activities are still ongoing. At the current stage, the main focus is on completing subsystem implementation and moving toward full system integration for the report submission workflow.

3.1.6 Expected Outcomes

The expected outcome of the project is a working prototype of a privacy-preserving, blockchain-based citizen reporting system for local governance.
The system is expected to achieve the following outcomes:

1. Anonymous civic reporting: Citizens can submit reports without directly revealing their real-world identity on-chain.

2. Tamper-evident records: Report metadata, hashes, and workflow states are recorded on a private blockchain.

3. Improved report reliability: AI-assisted moderation helps detect spam, abuse, and irrelevant submissions before blockchain anchoring.

4. Scalable storage: Large report content and media evidence are stored in IPFS, while blockchain stores only metadata and references.

5. Usable blockchain interaction: The relayer pattern removes the need for citizens to directly manage gas fees or blockchain transactions.

6. Modular deployment: The system is designed as a collection of independently deployable components.

7. Governance transparency: The blockchain provides an auditable record of report submission and status changes.

Through these outcomes, the prototype demonstrates the feasibility of combining blockchain, privacy-preserving identity mechanisms, off-chain storage, AI-assisted moderation, and web-based DApp interfaces to support transparent and accountable local governance reporting.

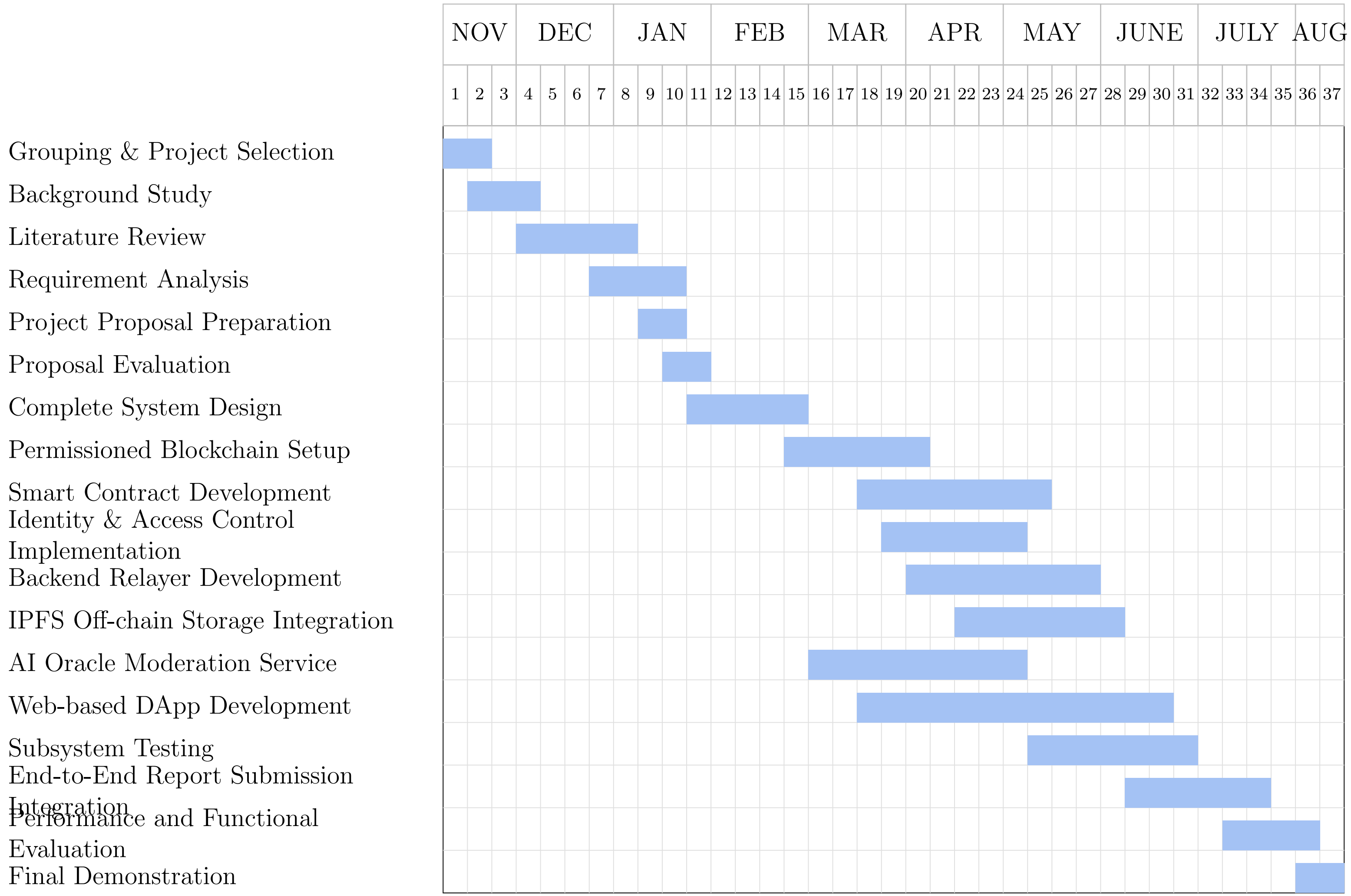

Figure 3.5: Project Timeline Gantt Chart

3.2 Implementation

3.2.1 Permissioned Blockchain Implementation

The core infrastructure of the proposed system is the private permissioned blockchain network. The implementation focused on setting up a secure and efficient environment tailored for local governance applications.

- Geth PoA Network Setup: A private permissioned Ethereum-compatible network was initialized using the Go Ethereum client. The network was configured to operate in proof-of-authority consensus mode, specifically using the Clique algorithm. This enabled faster block confirmation and lower energy consumption compared to traditional proof-of-work systems. A custom genesis file was created to define network parameters, including the initial set of authorized signers and the chain ID.

- Validator Node Configuration: Minimum of three full nodes were required to act as an validator. The 3 nodes are deployed in a personal vps infrastructure. Each node is configured with a unique cryptographic identity and their addresses were added to the list of authorized sealers within the consensus engine to manage transaction validation authority.

- Network Security Measures: To communicate with the blockchain network, secure RPC endpoint was established only for node1. Additionally, the network is configured to operate in a private subnet, further isolating it from external threats. This configuration allowed DApp and smart contracts to communicate with ledger enabling functions such as report submission, transaction querying and state synchronization.

3.2.2 Smart Contract Architecture and Logic

The core on-chain logic of the decentralized reporting framework is governed by the Reporting.sol smart contract. This contract is designed to be highly gas-efficient, acting primarily as a state machine and a cryptographic ledger rather than a data storage layer. The implementation is divided into three critical functional domains as The Report Contract structure, Ticket Validation, and State Transition Logic.

The Report Contract

The Report Contract is engineered using a Separation of Concerns principle. To mitigate the prohibitive gas costs associated with storing large multimedia files on the blockchain, the contract only stores essential metadata.

1. Data Structure: Each civic issue is stored in a Report struct containing a unique integer ID, the IPFS CID pointing to off-chain media, voting counters, the assigned authority address, and its current lifecycle state.

2. Role-Based Access Control (RBAC): Utilizing OpenZeppelin's AccessControl, the contract explicitly defines operational boundaries. The RELAYER_ROLE is restricted to the backend server, allowing it to submit reports and relay votes on behalf of citizens. The AUTHORITY_ROLE is granted to recognized government or NGO nodes, granting them permission to update an issue's status to solved or rejected.

Ticket Validation and Sybil Resistance

A fundamental challenge in anonymous decentralized systems is preventing Sybil attacks and duplicate submissions. This framework utilizes a dual-layer validation mechanism based on Zero-Knowledge Proof (Zero-Knowledge Proof (ZKP)) tickets.

1. Off-Chain Cryptographic Validation: Before a transaction reaches the blockchain, the backend relayer intercepts the payload. Using ethers.js, it verifies that the zkpTicketId possesses a valid cryptographic signature from the official Government Authority. It subsequently verifies the citizen's signature against the payload hash to ensure data integrity during transmission.

2. On-Chain Nullifier Tracking: Once validated off-chain, the zkpTicketId is passed to the smart contract as a submissionNullifier. The contract maintains a mapping of public submissionNullifiers. Upon successful report creation, this nullifier is marked as used. Any subsequent attempt to submit a report using the same ticket will systematically fail, cryptographically guaranteeing that one authorized ticket equates to exactly one unique report.

State Transition Logic

To enforce accountability and prevent unilateral dismissals of civic issues by authorities, the contract implements a strict Finite State Machine (Finite State Machine (FSM)). Every submitted report strictly adheres to an 8-stage lifecycle managed via an on-chain enum:

1. Pending Validation: Initial state awaiting community confirmation of legitimacy.

2. Community Rejected: Terminal state for spam or invalid issues.

3. Open: Validated by citizens and actively awaiting government action.

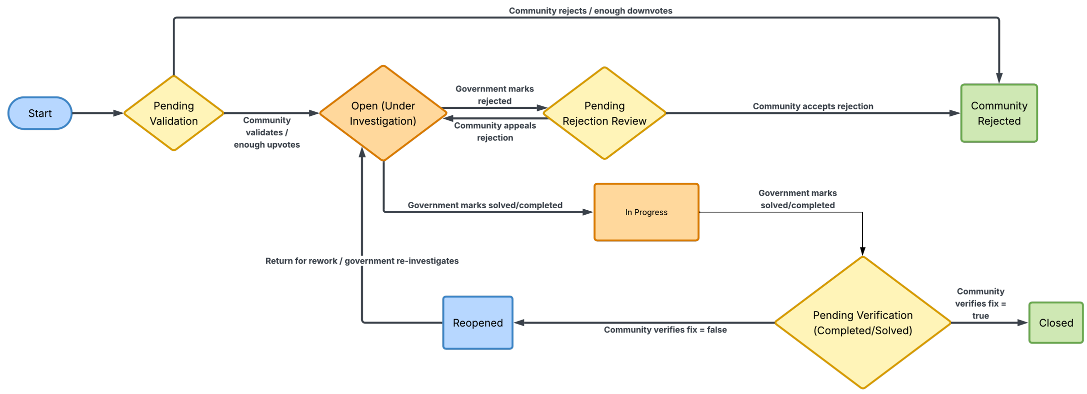

Figure 3.6: State transition diagram for the civic reporting system.

4. In Progress: The authority has officially acknowledged the issue and is actively executing physical repairs or resolution steps.

5. Pending Rejection Review: The authority dismissed the issue, triggering a community appeal phase.

6. Pending Verification: The authority marked the active work as completed, triggering a mandatory community verification vote.

7. Closed: Terminal state indicating a successfully verified resolution or an accepted rejection.

8. Reopened: The community rejected the authority's fix, forcing the issue back to the authority for rework.

A report must traverse these specific states driven by a combination of authority actions and community consensus thresholds.

1. Community-Driven Transitions: When a report is submitted, it enters the Pending Validation state. The transition to the Open state (making it visible for government action) only occurs automatically via the _changeStatus() internal function once community upvotes reach the predefined VALIDATION_THRESHOLD. Conversely, reaching the REJECTION_THRESHOLD moves it to the terminal Community Rejected state.

2. Authority-Driven Transitions: Authorities can only interact with reports in the Open, Reopened, or In Progress states. An authority calling startWork() shifts an Open or Reopened issue to the In Progress state, providing transparency to citizens that action is underway. From the In Progress state,

calling markAsSolved() shifts the report to Pending Verification. Additionally, calling rejectIssue() from the Open or In Progress states shifts it to Pending Rejection Review.

3. Consensus Verification: The FSM mandates that authority actions are never final until verified by the community. If an authority marks an issue as solved, the community must vote to verify the physical fix. Achieving the verification threshold transitions the report to the terminal Closed state. If the community rejects the fix, the state transitions to Reopened, forcing the authority to address the issue again.

3.2.3 Identity and Access Control Implementation

To satisfy the project's requirement for privacy-preserving accountability, an identity decoupling mechanism was successfully developed. This subsystem ensures that citizens can interact with the blockchain anonymously while remaining verified by an off-chain authority. The implementation was structured into three main components

- GovID Simulator: An off-chain simulated identity server was developed and deployed to a personal Virtual Private Server (Virtual Private Server (VPS)). This service acts as the trusted authority for verifying real-world identities. A dedicated API endpoint (/api/govid/verify-citizen) was implemented to securely authenticate users via their National Identity Number (govId) and a password, bridging the gap between traditional web authentication and decentralized access.

- Citizen Keypair generation: To completely decouple a user's real identity from their on-chain footprint, the identity server does not store or transmit blockchain addresses. Instead, upon successful authentication, the simulator generates and returns a secure, 256-bit citizenSeed. This cryptographic seed is then used by the client-side DApp to deterministically generate an anonymous public-private keypair, allowing the citizen to interact securely with smart contracts without exposing personal data.

- Ticket Batch Generation: To facilitate multiple secure, anonymous report submissions over time, the simulator implements a batch authorization protocol. During authentication, the server generates a structured ticketBatch (currently configured to issue 10 tickets per request). Each ticket contains a unique ticketId (a cryptographic hash) and a signature signed by the Identity Authority's private key. Citizens use these single-use tickets to authorize their transactions on-chain. The backend relayer validates the authority's signature on the ticket, granting access without ever knowing which specific citizen is submitting the report.

3.2.4 Backend Relayer Implementation

The backend relayer serves as a critical middleware component that bridges the frontend DApp with the blockchain network and external services. It is responsible for handling report submissions, managing interactions with IPFS for off-chain storage, and orchestrating AI-based content moderation before anchoring data on the blockchain.

The implementation utilizes the NestJS framework to provide a robust and highly modular RESTful API. The core workflow of the relayer executes the following deterministic pipeline for each incoming citizen report.

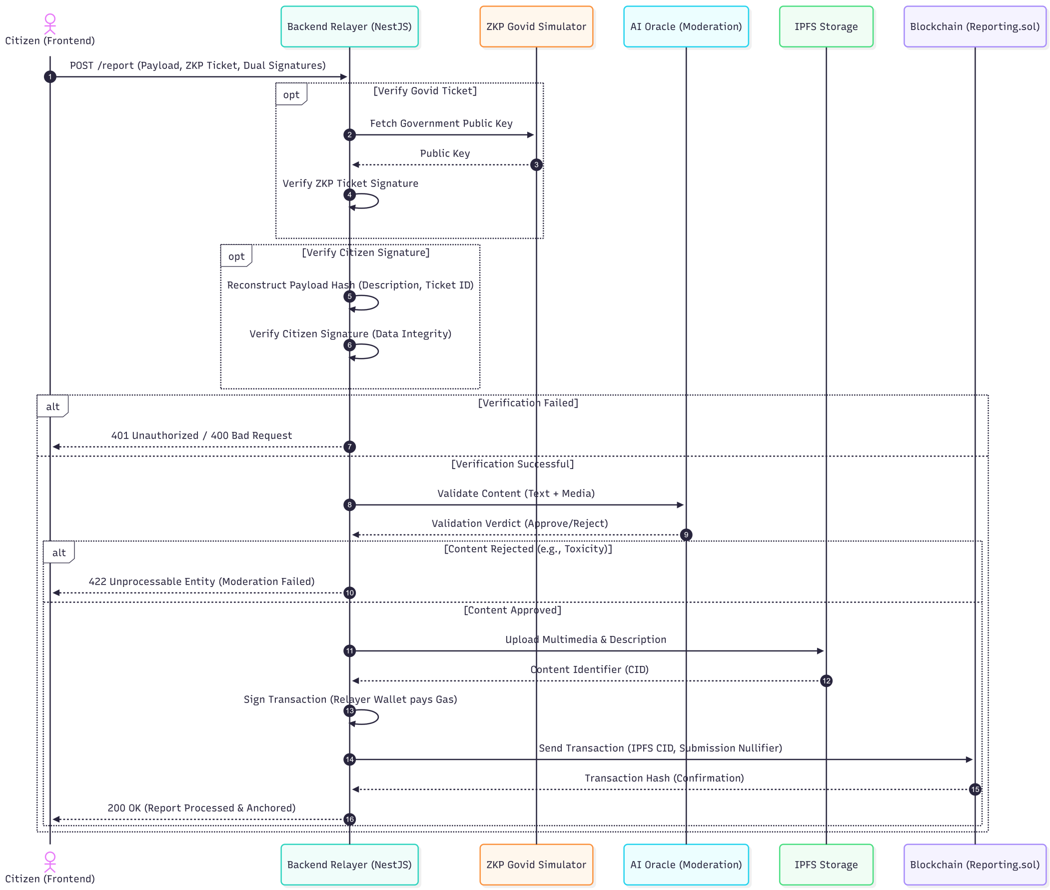

Figure 3.7: Backend relayer workflow architecture.

1. Cryptographic Verification: The relayer receives the payload via a POST /report endpoint. Utilizing the ethers.js library, it performs a strict dual-signature validation:

   (a) It verifies the ZKP Govid Simulator ticket signature against the dynamically fetched Government Authority public key to ensure the user is an authorized citizen.

   (b) It reconstructs the payload hash (binding the description and ticket ID) and verifies the citizen's signature to prevent data tampering in transit.

2. AI Content Moderation: To prevent on-chain spam, toxic immutability of blockchains and ensure platform integrity, the relayer routes the submitted text and media to the AI Oracle moderation service. This service acts as an automated filter, analyzing the content for appropriateness and returning a validation verdict before any storage operations occur.

3. Decentralized Storage (IPFS): To maintain a highly efficient and cost-effective blockchain state, heavy multimedia files and detailed text are offloaded to IPFS. The relayer manages the file buffers via multer, processes the upload, and retrieves an immutable CID.

4. Gasless Blockchain Submission: As the final step, the relayer formats a transaction containing the IPFS CID and the ZKP ticket (which serves as a Sybil-resistant submissionNullifier). The relayer signs and dispatches this transaction to the Reporting.sol smart contract. By paying the network gas fees from its own wallet, the relayer enables a completely frictionless, gasless experience for the end-user.

3.2.5 IPFS Off-chain Storage Integration

The system employs IPFS as the off-chain storage layer to manage large data objects, including citizen-submitted report images and structured report metadata. Rather than recording full file content on the blockchain, only the CID produced by IPFS is stored on-chain. The CID is a cryptographic hash uniquely representing the exact content of a file, such that any modification to the file produces an entirely different CID. This content-addressing mechanism ensures tamper-evident storage while significantly reducing on-chain data overhead and improving overall system scalability.

To demonstrate the decentralized nature of IPFS within the prototype environment, three IPFS nodes are deployed across three separate machines and connected as peers within a private controlled network. When a file is uploaded and pinned on the primary node, the two peer nodes retrieve and independently pin the same content, ensuring redundant availability across all three machines. This peer network operates entirely within the controlled laboratory environment and does not connect to the public IPFS network, which is consistent with the defined project scope.

The off-chain storage integration is implemented as a dedicated backend service positioned between the DApp frontend and the IPFS node cluster, structured using a layered architecture of route definitions, controller components, utility modules, and middleware layers. A content moderation step is executed before any file is committed to IPFS storage, where every submission passes through an AI-assisted moderation module and only approved content is uploaded and assigned a CID. At the current stage of development, the moderation module has been integrated into the service architecture and is prepared to connect to the AI moderation service once that component is completed.

The upload workflow follows a two-stage sequence. In the first stage, the submitted image passes through moderation, is uploaded to the primary IPFS node, pinned across peer nodes, and its CID is returned. In the second stage, the complete report metadata comprising the title, description, category, geographic location, pseudonymous reporter identifier, and the image CID is packaged as a structured JSON object and uploaded to IPFS, producing a metadata CID. The DApp then forwards only this metadata CID to the smart contract, which records it permanently on the blockchain as an immutable on-chain anchor. During retrieval, the metadata CID is read from the blockchain and used to fetch the full report data from IPFS, with the embedded image CID used separately to retrieve the associated image.

The system further incorporates a CID verification mechanism to confirm that a CID is resolvable before it is submitted to the blockchain, and a pin management mechanism to handle content removal where necessary. Removing a pin from the local nodes allows garbage collection to reclaim storage while the on-chain CID record is preserved as an immutable audit trail, directly addressing the toxic immutability challenge identified in the literature.

3.2.6 AI-based Moderation System Implementation

The AI moderation subsystem is designed as an off-chain oracle network. The moderation process is performed off-chain and the result is returened to the reporting workflow. The oracle network consists of three independent oracle workers and one aggregator. The aggregator forwards the moderation request to all oracle workers, collects their decisions, applies a voting rule, signs the final decision, and returns the result to the relayer. This architecture is demonstrated in Figure 3.8.

The oracle 1 focuses on detecting toxic, abusive, threaening, or harmful language. It analyzes the submitted report using a predefined moderation strategy and re-

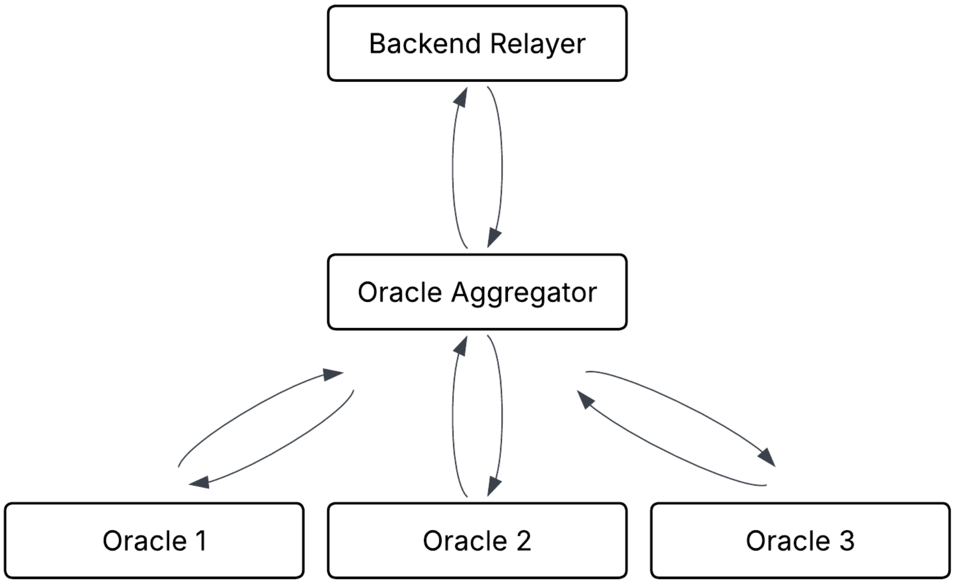

Figure 3.8: Oracle network architecture for AI moderation.

turns a vote. The oracle 2 focuses on spam detection. It checks for promotional content, repeated characters, and extremely short descriptions. The third oracle focuses on civic relevance. It checks whether the report is related to local governance issues such as roads, garbage, water, drainage, public infrastructure, and community facilities.
The oracle aggregator combines the decisions of the three oracle workers. Each oracle returns either ACCEPT or REJECT. The current aggregation logic uses a two out of three (2/3) majority rule. The final moderation result includes the final decision, confidence score, report hash, oracle votes, decision hash, and aggregator signature.
To ensure integrity of the moderation results, several mechanisms are used. The aggregator computes a hash of the submitted report payload. The final moderation result and oracle votes are combined and hashed to crate the decision hash. Also, the aggregator sign the decision hash using its private key.

3.2.7 Web-based DApp Development

The decentralized application (DApp) serves as the primary gateway for citizens to interact with the local governance system. It is built using a modern web stack comprising Next.js and React for the frontend interface, Tailwind CSS for responsive styling, and Ethers.js for cryptographic operations. The architecture is designed to abstract blockchain complexities away from the end-user while maintaining strict privacy and security guarantees.

User Authentication Interface

The authentication interface is designed to verify citizen eligibility without compromising personal privacy.

- **GovID Verification Integration:** The interface securely collects the citizen's GovID credentials and interfaces with an external GovID Simulator API. This abstracts the process of verifying a citizen's real-world identity against a centralized registry.
- **Cryptographic Credential Generation:** Upon successful identity verification, the system retrieves a batch of ZKP tickets and a unique 'citizenSeed'.
- **Local State Management:** Using the 'ethers.js' library, a localized Citizen Wallet is derived from the seed. A custom React Context (`CitizenContext`) is utilized to manage the session state, ensuring that the generated private keys and ZKP tickets remain strictly on the client-side device and are never exposed to external servers. The UI incorporates dynamic feedback mechanisms to indicate the generation of cryptographic proofs to the user.

Report Submission Interface

The report submission module allows authenticated citizens to report civic issues anonymously.

- **Form Design and Data Handling:** Developed with a mobile-first approach, the form captures issue descriptions and optional photographic evidence using standard `FormData` structures to support multipart uploads.

- **Anonymous Cryptographic Signing:** To guarantee anonymity and prevent spam, each submission consumes a single ZKP ticket from the user's allocated batch. The payload (consisting of the issue description and the ZKP ticket ID) is hashed using Keccak-256 and subsequently signed using the citizen's locally stored ECDSA private key.

- **Meta-Transaction Relaying:** Instead of submitting transactions directly to the blockchain and incurring gas fees, the frontend dispatches the signed payload, image, and ZKP metadata to a NestJS Backend Relayer. The relayer handles the intermediate steps: uploading the evidence to IPFS, requesting AI moderation, and finally submitting the transaction to the Geth blockchain network on behalf of the user.

Blockchain Query Interface

The query interface acts as a community feed where citizens can view, filter, and track the status of reported issues.

- **Mobile-First Feed Architecture:** Implemented as a single-column, scrollable interface inspired by modern social media platforms (e.g., Reddit). This design choice prioritizes readability and user engagement on mobile devices.

- **Dynamic Issue Cards:** Data retrieved from the blockchain is formatted into interactive cards. Each card displays critical metadata including categorical status badges (e.g., PENDING VALIDATION, OPEN, RESOLVED), community upvotes, truncated descriptions, timestamps, and image thumbnails.

- **State Management and Navigation:** The interface employs React state hooks to allow real-time filtering of issues by category (e.g., Infrastructure, Parks & Rec, Safety). Furthermore, a persistent Floating Action Button (FAB) is implemented to provide users with seamless access to the report submission workflow at any time.

Chapter 4

Testing, Experimentation, and Validation

This chapter outlines the proposed methodology for the testing, experimentation, and validation of the decentralized reporting framework. Given the distributed nature of the system—which integrates off-chain cryptographic services, AI moderation, and on-chain smart contracts—a rigorous experimental setup will be established to evaluate performance, security, and functional correctness.

4.1 Experimental Setup

To accurately simulate a real-world deployment of the permissioned blockchain and its ancillary microservices, the experimental environment has been hosted on a cloud-based infrastructure utilizing containerization technologies.

4.1.1 Hardware Infrastructure

The testing environment was deployed across two identical VPS instances to simulate a distributed network architecture and evaluate inter-service communication latency. The hardware specifications for each VPS are as follows:

1. Compute: 4 vCPU Cores
2. Memory: 8 GB RAM
3. Storage: 75 GB SSD
4. Network: High-bandwidth cloud uplink

Server 1 acts as the primary operational node, hosting the core citizen-facing applications, middleware, and the blockchain ledger. Server 2 is strictly dedicated to hosting the computationally intensive AI moderation ensemble and the IPFS storage layer. This isolation prevents resource monopolization and ensures accurate performance profiling of the blockchain consensus mechanism.

4.1.2 Software and Deployment Environment

To ensure reproducibility and consistency across test iterations, the entire software stack was containerized using Docker.

1. Orchestration and CI/CD: Deployment and lifecycle management of the containers are handled by Dokploy, a self-hosted Platform-as-a-Service (PaaS) solution installed on both servers.

2. Automated Deployment: Continuous Integration and Continuous Deployment (CI/CD) pipelines were established via GitHub webhooks. Dokploy was configured to monitor the develop branch of the project repository. Upon receiving a trigger, it automatically pulls the latest source code, builds the respective Dockerfiles for the microservices, and deploys the updated containers.

4.1.3 Blockchain Configuration (Server 1)

The on-chain environment consists of a private, permissioned PoA blockchain.

1. Client Software: The network was initialized using Go-Ethereum (Geth) version 1.13.15.

2. Consensus Engine: The Clique PoA consensus algorithm was utilized to ensure deterministic block times and eliminate the computational overhead of PoW, which closely aligns with the operational model of a government-backed consortium chain.

3. Node Topology: A 3-node cluster was deployed via a unified docker-compose.yml configuration. These nodes serve as the designated Authority (Sealer) nodes representing various governance entities.

4.1.4 Application and Services Topology

The testing setup evaluates the interaction between the following primary subsystems distributed across the two servers.

Deployed on Server 1 (Core Operations):

1. ZKP-GovID Simulator: A Node.js service simulating the issuance of government-backed ZKP tickets.
2. Frontend DApp: A Next.js application providing the citizen interface.
3. Backend Relayer: A NestJS-based middleware component responsible for off-chain cryptographic validation, routing, and gasless blockchain transaction submission.
4. Smart Contracts: Compiled using the Hardhat development environment utilizing Solidity version 0.8.20.

Deployed on Server 2 (Storage and AI Oracle):

1. AI Moderation Ensemble: A multi-node AI oracle architecture designed for robust spam and content filtering. This consists of three distinct AI algorithm nodes running concurrently to analyze incoming media and text. These nodes are governed by a central Aggregator Node that acts as the single entry point and determines the final moderation verdict based on algorithmic consensus.
2. IPFS Node: A dedicated IPFS node that will be deployed in the subsequent testing phase to handle the distributed, off-chain storage of all civic report multimedia (images) and heavy metadata, returning immutable CIDs to the backend relayer.

4.2 Testing Methodology

To ensure the reliability, security, and performance of the decentralized reporting framework, a comprehensive, multi-tiered testing methodology will be employed. This approach systematically validates the architecture from isolated components up to the complete end-to-end user workflow.

4.2.1 Subsystem Testing

Subsystem testing will focus on isolating individual architectural components to verify their internal logic, cryptographic integrity, and algorithmic correctness prior to network integration.

1. Smart Contract Validation: The core reporting smart contract will undergo rigorous unit testing using the Hardhat development framework. Automated tests will be written to validate the logic of the FSM, ensuring that community voting thresholds trigger correct status changes and that role-based access controls prevent unauthorized authority actions.

2. Backend Relayer Verification: The middleware will be tested using a dedicated testing suite. The primary focus will be validating the cryptographic verification algorithms, ensuring that both the ZKP tickets from the simulator and the citizen signatures are correctly authenticated or systematically rejected when malformed.

3. AI Oracle Evaluation: The AI moderation ensemble will be tested in isolation using a curated dataset of synthetic civic reports. This dataset will include a mixture of valid public issues, explicit spam, and manipulated imagery to evaluate the accuracy, false-positive rate, and consensus latency of the multi-node aggregator.

4.2.2 Integration Testing

Once individual subsystems are validated, integration testing will be conducted to evaluate the communication interfaces, network latency, and data flow between the distributed microservices across the two virtual private servers.

1. Relayer and Off-Chain Services: Tests will verify the backend relayer's ability to seamlessly orchestrate the initial validation pipeline. This includes measuring the successful routing of payloads to the AI Oracle ensemble, the handling of timeout exceptions, and the successful uploading of multimedia to the IPFS node to retrieve immutable CIDs.

2. Relayer and Blockchain Interface: Integration scripts will simulate the relayer dispatching transactions to the private PoA blockchain network. This phase will verify the gasless submission mechanics, ensuring the relayer accurately wraps the CIDs and ticket nullifiers into valid transactions that are accepted by the network nodes.

3. Frontend to Middleware Communication: API integration tests will confirm that the decentralized application securely formats and transmits the cryptographic payloads, including the user description and attached media, to the relayer over encrypted channels.

4.2.3 Functional Testing

Functional testing will evaluate the entire integrated system against the defined operational requirements from an end-user perspective, simulating both real-world usage and adversarial conditions.

1. End-to-End User Journeys: Automated scripts will simulate complete, valid report lifecycles. This entails a simulated citizen generating a ticket, submitting a valid issue via the frontend, passing AI moderation, anchoring on the blockchain, and successfully progressing through the community validation and authority resolution states.

2. Sybil Resistance and Replay Attack Mitigation: The framework will be subjected to adversarial functional tests. Scripts will intentionally attempt to submit multiple reports using identical ticket nullifiers to verify that the smart contract successfully rejects duplicate entries and maintains the one-ticket-one-vote invariant.

3. Cryptographic Forgery Attempts: The system will be tested by injecting payloads with manipulated descriptions or mismatched signatures. This will ensure the backend relayer systematically drops tampered requests before they can interact with the off-chain storage or consume blockchain resources.

4.2.4 Blockchain Performance Testing

To quantitatively evaluate the viability and scalability of the proposed decentralized reporting framework, a series of performance benchmarking experiments will be conducted on the 3-node Proof-of-Authority network. These stress tests will be executed using automated benchmarking tools, such as Hyperledger Caliper or custom Node.js scripting, to simulate concurrent user interactions. The evaluation will focus on three primary Key Performance Indicators as transaction throughput, transaction latency, and block confirmation times.

Transaction Throughput Testing

The throughput experiment will measure the maximum capacity of the blockchain network to process high volumes of simultaneous civic reports.

1. Experiment Design: The testing suite will generate a progressively increasing load of concurrent report submissions and voting transactions sent to the backend relayer. This mimics a real-world scenario of mass reporting during a localized civic emergency, such as a natural disaster.

2. Significance: High throughput is essential to prevent network congestion, avoid memory pool overflows, and ensure the relayer does not become a bottleneck during peak usage.

3. Metrics and Parameters: The primary metric is Transactions Per Second, calculated by dividing the total number of successfully committed transactions by the total duration of the stress test. Secondary parameters include the transaction success-to-failure ratio and the CPU utilization on Server 1.

4. Thresholds and State-of-the-Art Comparison: Public mainnet blockchains like Ethereum traditionally process between 15 and 30 Transactions Per Second. The benchmark for this permissioned Proof-of-Authority configuration is set at a minimum threshold of 100 Transactions Per Second. Achieving this will demonstrate a highly scalable environment that significantly outperforms traditional public ledgers for civic applications.

Transaction Latency Testing

The latency experiments will evaluate the responsiveness of the reporting system from the perspective of the backend relayer and the end-user.

1. Experiment Design: High-precision timestamps will be recorded at the exact moment the relayer dispatches the signed transaction to the blockchain and compared against the timestamp of the finalized block that incorporates that specific transaction.

2. Significance: Low transaction latency ensures a seamless user experience. Extended delays could lead to timeout errors within the middleware interface and cause desynchronization between the off-chain IPFS storage and the on-chain smart contract state.

3. Metrics and Parameters: The core metrics evaluated will be the Average Transaction Latency measured in seconds, and the 95th percentile latency to account for network outliers under heavy load conditions.

4. Thresholds and State-of-the-Art Comparison: While public Proof-of-Stake networks currently target latency periods of 12 seconds or more, the configured Proof-of-Authority consensus mechanism eliminates complex cryptographic puzzles. The expected benchmark for this project is an average latency of under 5 seconds per transaction, representing a substantial optimization over standard public blockchain networks.

Block Confirmation and Finality Testing

This phase of testing will analyze the deterministic nature of the Clique consensus engine across the three designated authority nodes.

1. Experiment Design: Network monitoring tools will track the block propagation times across the node topology and measure the block generation intervals during both idle periods and extreme stress-testing phases.

2. Significance: Rapid and deterministic block confirmation is critical for the FSM implemented in the reporting contract. It ensures that community votes and authority state transitions are registered instantly without the risk of chain reorganizations, forks, or orphaned blocks.

3. Metrics and Parameters: The primary parameters are Average Block Time measured in seconds and Time to Finality, which measures the duration required for a transaction to become mathematically irreversible.

4. Thresholds and State-of-the-Art Comparison: Traditional probabilistic finality in legacy networks can take minutes to resolve. For this 3-node Proof-of-Authority setup, the block time parameter is explicitly configured in the genesis block to a fixed interval of 5 seconds. Consequently, the benchmark for absolute finality is expected to be achieved within 1 to 2 blocks, or 5 to 10 seconds. This matches the state-of-the-art performance observed in modern enterprise consortium blockchains.

4.2.5 Smart Contract Validation

To ensure the logical correctness, security, and gas efficiency of the decentralized reporting framework, a comprehensive validation suite will be executed against the core Solidity smart contract. This testing phase will utilize the Hardhat development environment paired with the Chai assertion library to simulate complex on-chain interactions. The validation process is categorized into three critical domains as ticket validation, report creation logic, and state transition integrity.

Ticket Validation and Sybil Resistance

This phase will empirically test the cryptographic nullifier tracking system to ensure the network is impervious to Sybil attacks and duplicate reporting.

1. Experiment Design: Automated test scripts will simulate the submission of civic reports using valid, backend-signed ZKP tickets. Immediately following a successful submission, the script will intentionally attempt a replay attack by submitting a secondary report utilizing the exact same ticket nullifier. Additional tests will attempt to inject malformed or mathematically invalid nullifier hashes.

2. Significance: Validating the nullifier mapping is critical to preserving the one-citizen-one-vote invariant. If a malicious actor could bypass this check, they could artificially inflate the severity of an issue or spam the government authority with duplicate tasks.

3. Metrics and Parameters: The primary metric evaluated will be the Transaction Revert Rate under adversarial conditions, alongside the specific Gas Cost required to execute the nullifier validation logic.

4. Thresholds and Benchmarks: The acceptable benchmark for this mechanism is a strict 100 percent transaction revert rate for all duplicate or malformed ticket submissions. Furthermore, the gas overhead for checking the nullifier mapping must remain highly optimized to ensure the relayer is not subjected to excessive network fees.

Report Creation and Access Control

These experiments will validate the initialization of civic issues on the blockchain, ensuring that data is stored efficiently and that only authorized entities can write to the ledger.

1. Experiment Design: The test suite will generate transactions attempting to call the report creation function from various network addresses. One address will be properly provisioned with the Relayer Role, while the others will represent unauthorized standard user wallets. The test will verify that only the relayer can anchor the IPFS CID and initial voting metadata to the chain.

2. Significance: This verifies the Separation of Concerns architecture, ensuring the smart contract acts exclusively as a secure state machine rather than a vulnerable, open-access database.

3. Metrics and Parameters: The parameters assessed include RBAC execution accuracy and the Initial State Mapping accuracy. The computational cost will be measured in total execution gas.

4. Thresholds and Benchmarks: Any creation attempt by an address lacking the designated relayer permissions must revert immediately.

State Transition and FSM Logic

The most rigorous testing phase will evaluate the strict 8-stage lifecycle of a civic report to guarantee that government authorities and citizens interact within predefined boundaries.

1. Experiment Design: A simulated environment will recreate the complete lifecycle of multiple reports. Tests will incrementally cast community votes to verify that an issue automatically transitions from Pending Validation to Open upon reaching the required mathematical threshold. Subsequently, simulated authority nodes will invoke state-changing functions such as starting work, marking as solved, or rejecting the issue. Finally, the suite will attempt unauthorized state bypasses, such as an authority attempting to manually force an issue into the Closed state without the mandatory community Pending Verification phase.

2. Significance: The FSM is the core accountability mechanism of the project. Validating this logic ensures that authorities cannot unilaterally dismiss public issues and that community consensus dictates the final resolution of every report.

3. Metrics and Parameters: The primary metric is Code Coverage, specifically targeting statement, branch, and function coverage within the smart contract. Secondary metrics include State Transition Accuracy under edge-case voting scenarios.

4. Thresholds and Benchmarks: The testing benchmark mandates 100 percent branch and statement coverage for all state transition functions. The system must yield zero successful unauthorized state bypasses, enforcing absolute adherence to the 8-stage consensus lifecycle.

4.3 AI Moderation Testing

AI moderation testing evaluates whether the moderation subsystem correctly classifies submitted report content as acceptable or unacceptable. At the current stage, the implemented AI moderation service focuses mainly on text-based moderation. Image moderation is planned as a future extension.
The objectives of AI moderation testing are:

1. to verify that the oracle aggregator API is accessible

2. to verify that all three oracle workers respond correctly

3. to test valid civic report acceptance

4. to test spam and abusive content rejection

5. to verify majority voting behavior

6. to verify generation of report hash, decision hash, and signature

The AI moderation service is deployed on the server 2 using Dokploy and Docker containers. The deployment includes:

1. Government Oracle container
2. International Oracle container
3. NGO Oracle container
4. Oracle Aggregator container

The backend relayer communicates only with the aggregator API. The aggregator communicates with the three oracle workers using internal container networking.

4.3.1 API Availability Test

The test cases used for testing moderation API availability are shown in the Table 4.1

Table 4.1: AI moderation API availability tests

Test ID	Test	Expected Result	Status
AI-T01	Call aggregator health endpoint	Service returns running status	Passed
AI-T02	Check oracle worker availability	Aggregator identifies available oracle workers	Passed
AI-T03	Send request without API key	Request rejected	Passed
AI-T04	Send request with valid API key	Request processed successfully	Passed

4.3.2 Moderation Classification Tests

The test cases used for testing moderation classification are shown in Table 4.2

4.3.3 Oracle Voting Validation

The oracle moderation system uses a three-node voting mechanism. Each oracle returns either one of the votes, `ACCEPT` or `REJECT`.
The aggregator applies majority-based decision logic.

Table 4.2: AI moderation classification tests

Test ID	Input Type	Example Input	Expected Decision
AI-T05	Valid civic report	"There is a large pothole near the public school."	ACCEPT
AI-T06	Spam content	"BUY NOW FREE MONEY CLICK HERE."	REJECT
AI-T07	Toxic content	Text containing abusive or threatening phrases	REJECT
AI-T08	Empty report	Empty text field	Reject request / validation error

Table 4.3: Oracle voting validation

Oracle Votes	Expected Final Decision
ACCEPT, ACCEPT, ACCEPT	ACCEPT
ACCEPT, ACCEPT, REJECT	ACCEPT
REJECT, REJECT, ACCEPT	REJECT
REJECT, REJECT, REJECT	REJECT

4.3.4 Response Structure Validation

Each successful oracle response should include the fields shown in Table 4.4.

Table 4.4: Oracle aggregator response fields

Field	Purpose
`final_decision`	Final moderation result
`final_confidence`	Aggregated confidence value
`report_hash`	Hash of submitted report payload
`decision_hash`	Hash of final moderation decision
`oracle_votes`	Individual oracle outputs
`aggregator_signature`	Signature of decision hash

The presence of `report_hash`, `decision_hash`, and `aggregator_signature` supports auditability and tamper-evidence in the moderation process.

The AI content moderation service was successfully deployed as separate containers on a VPS. Preliminary API testing confirms that the aggregator can receive moderation requests, communicate with the oracle workers, collect votes, and return an aggregated decision. This validates the feasibility of the decentralized oracle moderation approach at prototype level.

4.4 IPFS Storage Testing

IPFS serves as the off-chain storage layer of the proposed system. Since storing large data such as images and full report metadata directly on the blockchain is expensive and impractical, the system uploads these files to IPFS and records only the resulting CID on-chain. The CID is a cryptographic hash of the file content, meaning if the file changes in any way, the CID changes as well. This design ensures data integrity without burdening the blockchain with large files.
At the current stage of development, the IPFS service layer has been successfully implemented and the connection between the service and the local IPFS node has been established and verified. A health verification test was conducted to confirm that the IPFS service is running correctly and that it communicates properly with the underlying IPFS daemon. The test confirmed a healthy status along with accurate node version information, establishing that the foundation of the off-chain storage layer is stable and ready for further integration.
However, since the DApp frontend, the AI content moderation module, and the blockchain layer have not yet been completed, the full testing of the IPFS storage integration is still in progress. The remaining testing activities will be carried out progressively as the dependent components are completed. Upload performance testing will be conducted once the DApp frontend is available to submit real citizen reports through the system. Files of different sizes, including small JSON metadata objects and larger image files, will be uploaded through the IPFS service layer and the time taken for each upload will be recorded and analyzed to assess the responsiveness of the system under realistic conditions. Retrieval integrity testing will be performed to confirm that files downloaded from IPFS remain identical to the originals. After uploading a file and obtaining its CID, the same file will be retrieved using that CID and compared against the original content to verify that nothing was lost or altered during storage and retrieval. CID validation testing will be carried out to ensure that the generated CID accurately represents the corresponding file content. For each uploaded file, the CID produced by IPFS will be recorded, and after retrieval, the CID will be recalculated from the downloaded content and compared against the original value to confirm that the content-addressing mechanism functions correctly. AI-assisted content moderation integration testing will be performed once the moderation service is

connected to the IPFS layer. These tests will verify that inappropriate images and text are correctly identified and rejected before being permanently stored, ensuring that only approved content reaches the IPFS node and subsequently the blockchain. End-to-end integration testing between the IPFS storage layer and the deployed smart contracts will be conducted once the blockchain layer is completed. These tests will verify that CIDs recorded on-chain correctly resolve back to their full report metadata and images when fetched through the IPFS retrieval layer, confirming that the complete off-chain and on-chain data flow operates as intended.

4.5 End-to-End Integration Testing

To ensure the seamless operation of the entire platform, end-to-end integration testing will be conducted by simulating the complete lifecycle of a civic issue report. This testing phase will verify the interoperability between the frontend DApp, the backend relayer, the IPFS storage network, the AI moderation oracle, and the underlying permissioned blockchain.
The end-to-end testing workflow will be executed as follows:

1. Citizen Authentication and Key Generation: The test will commence with a simulated citizen authenticating via the external GovID simulator. Upon successful verification, the system is expected to generate a batch of ZKP tickets and derive the local wallet, thereby validating the correct functioning of the identity decoupling mechanism.

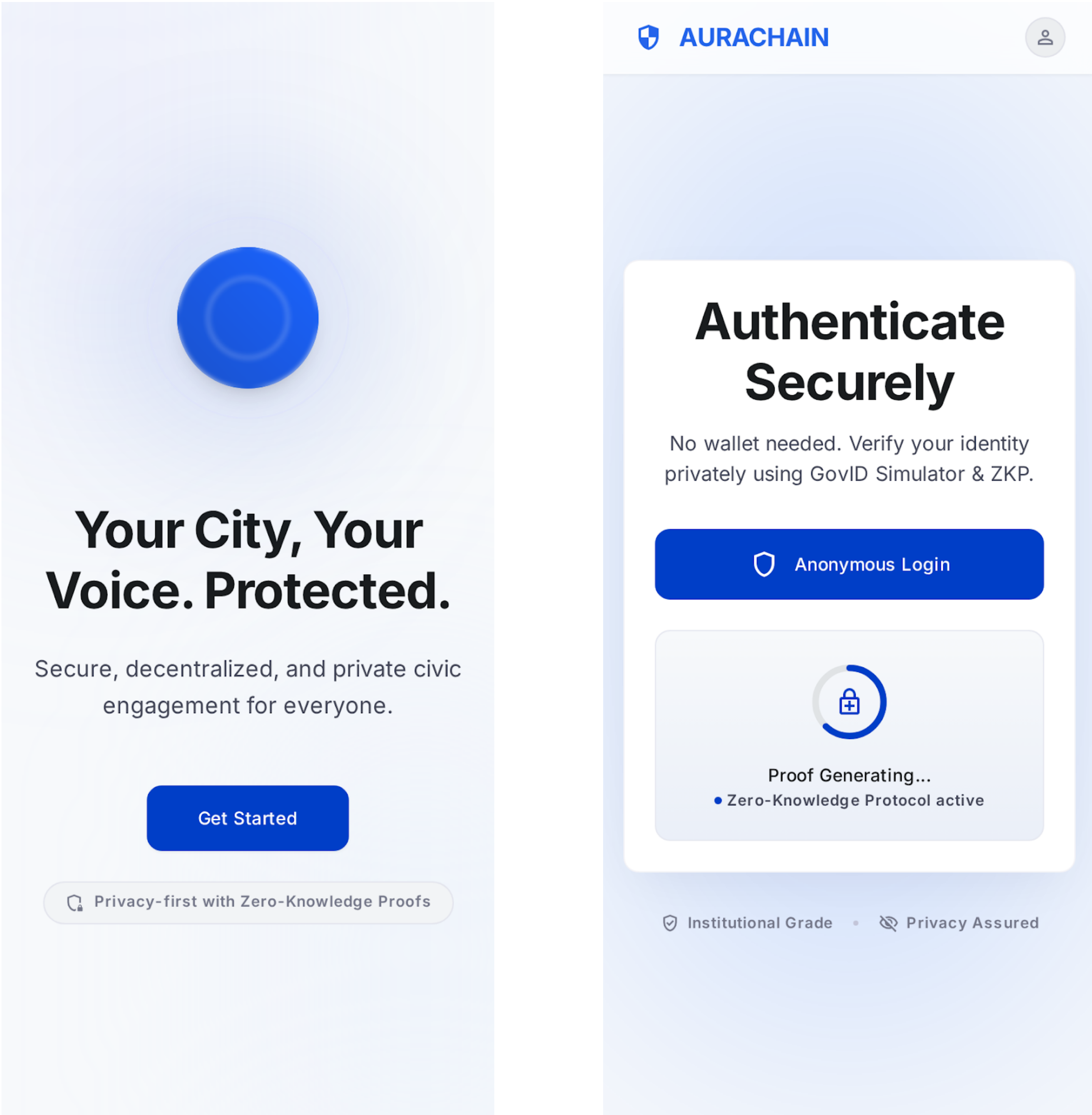

Figure 4.1: Citizen Authentication Interfaces

2. Report Submission and Cryptographic Signing: The authenticated user will submit a test report containing a text description (e.g., "Pothole on Main Street") and a sample image. The DApp will be expected to consume a ZKP ticket, hash the payload, and sign it using the local private key before transmitting the multipart form data to the backend relayer.

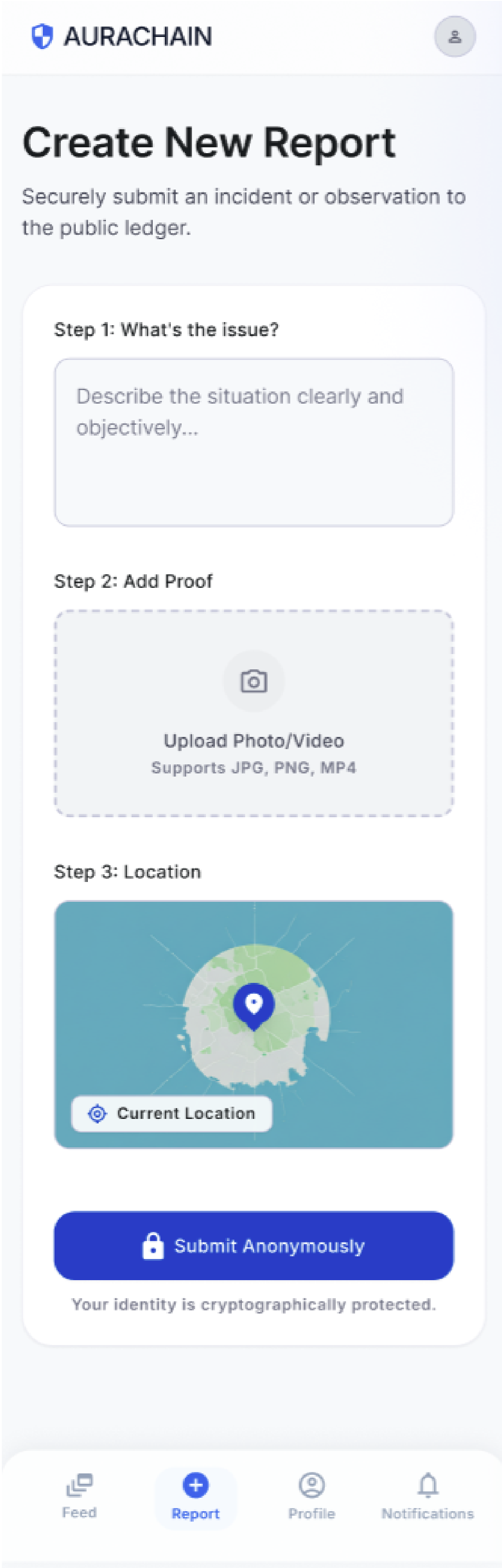

Figure 4.2: Report Submission Form

3. Off-chain Storage and AI Moderation Routing: Upon receiving the payload, the backend relayer will process the submitted data. The visual evidence will be routed to the IPFS integration module, and a CID will be generated. Simultaneously, the text and image data will be evaluated by the AI moderation service to verify that the report is classified as legitimate, non-abusive content before proceeding to blockchain submission.

4. Blockchain Recording via Meta-Transaction: Following AI approval, the backend relayer will format the data, including the IPFS CID, issue description, and the cryptographic proofs. Then submit a meta-transaction to the smart contract on the Geth PoA network. The transaction is expected to be verified by the validator nodes, securely anchoring the report to the immutable ledger without requiring the citizen to pay any gas fees.

5. Frontend Verification: Finally, the DApp's query interface will be refreshed to retrieve the latest blockchain state. The expected outcome is that the newly submitted report will appear in the community feed with its corresponding status badge, upvote count, and image thumbnail, confirming that the entire decentralized pipeline has executed correctly.

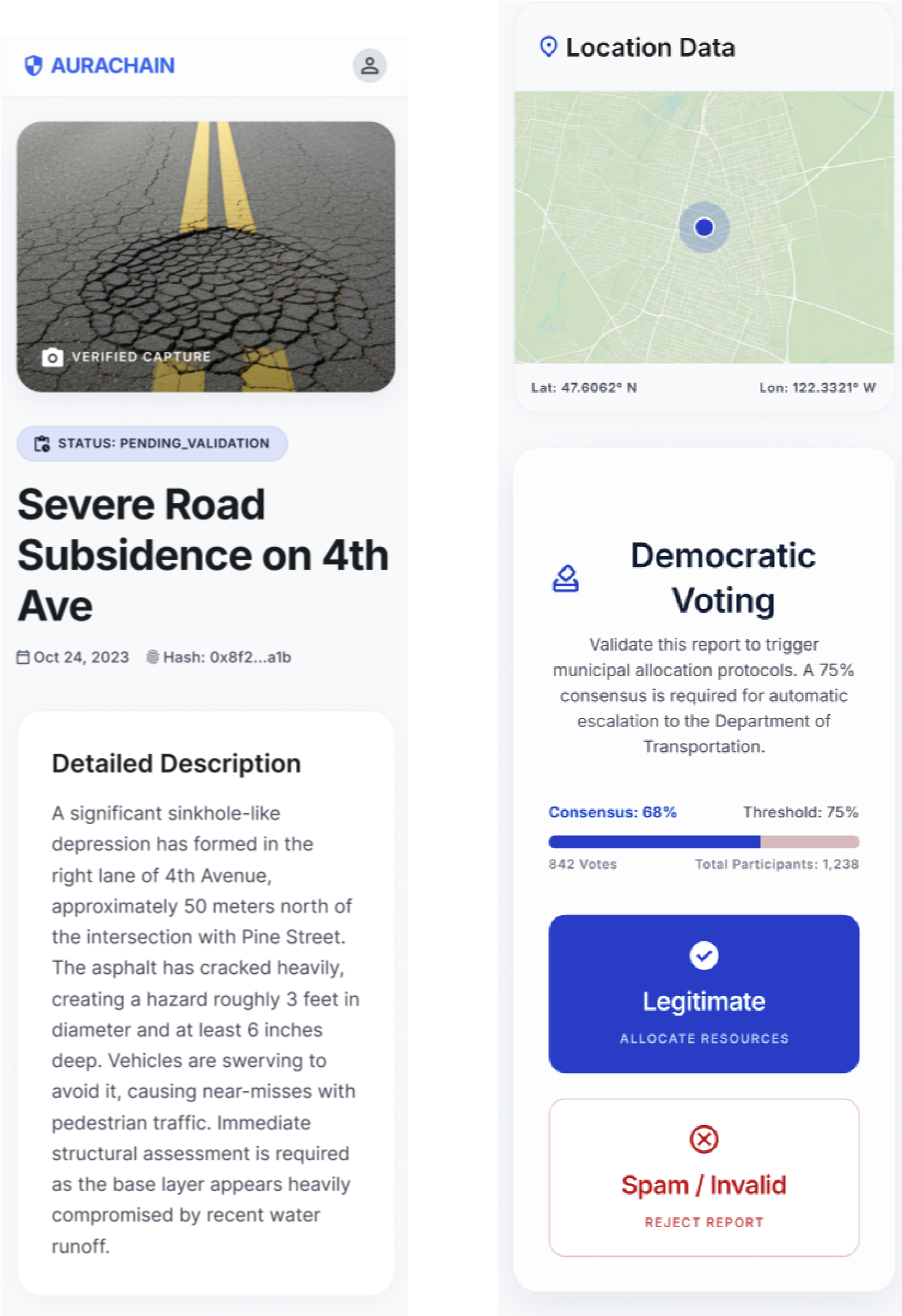

Figure 4.3: Views of the Submitted Report in the Community Feed

The successful execution of this simulated workflow will validate the system architecture and confirm that the individual components interact correctly to provide a secure, anonymous, and user-friendly civic reporting platform.

4.6 Comparison with Existing Systems

To demonstrate the advantages and novel contributions of the proposed community-assisted reporting framework, it is essential to compare its architecture and features against existing solutions in the domain of e-governance and citizen reporting.

4.6.1 Centralized Reporting Platforms

Traditional crowdsourced reporting systems, such as FixMyStreet and SeeClickFix, have successfully established user-friendly interfaces with intuitive reporting workflows and clear feedback loops. However, as identified in the literature, these platforms rely entirely on centralized server architectures. This centralization creates single points of failure and makes the systems susceptible to data tampering and unauthorized modifications. Furthermore, centralized platforms often require mandatory identity disclosure, which can discourage citizen participation, especially when reporting sensitive issues or criticizing local authorities. In contrast, the proposed system leverages a permissioned blockchain to ensure data immutability and absolute transparency. By employing ZKP and identity decoupling, the proposed framework guarantees citizen anonymity while maintaining mathematical accountability, a feature completely absent in conventional centralized systems.

4.6.2 Traditional Content Moderation

Existing governance platforms typically rely on manual content moderation by human administrators to filter out spam or abusive content. While effective for small-scale operations, manual moderation introduces significant transaction latency, high operational costs, and the potential for human bias. Most importantly, it reintroduces centralized control over what public information is allowed to be published. The proposed system addresses these limitations by integrating an AI-based off-chain moderation oracle. By automating the filtering of textual and visual content using machine learning models before the data is permanently recorded on-chain, the system achieves scalable, unbiased, and rapid moderation while preserving the decentralized ethos of the platform.

4.6.3 Public Blockchain Approaches

While some theoretical governance models propose using public, permissionless blockchains (such as the Ethereum mainnet) to achieve maximum decentralization, these architectures are poorly suited for frequent local governance interactions. Public blockchains suffer from high transaction latency, limited throughput, and unpredictable, fluctuating gas fees that the user must pay. The proposed framework overcomes these usability challenges by utilizing a private, permissioned blockchain running a PoA consensus mechanism. This architecture provides predictable, high-performance transaction processing. Additionally, it eliminates the financial burden on citizens through the use of a backend relayer network (meta-transactions), offering a seamless, "gasless" experience.

4.6.4 Domain-Specific Blockchain Solutions

Recent literature highlights several blockchain-based complaint management systems, but these are primarily tailored for highly specific domains such as law enforcement (e.g., police complaint registries). These systems often focus narrowly on administrative efficiency, restrict access entirely to authorized personnel, and frequently lack robust mechanisms for handling large multimedia evidence. Unlike these isolated solutions, the proposed framework introduces a generalized, community-assisted approach. It actively encourages public participation through opinion polling and community upvoting. Furthermore, it efficiently manages large multimedia files (e.g., images of infrastructure damage) through decentralized off-chain storage via IPFS, seamlessly linking cryptographic hashes back to the immutable blockchain ledger.

Chapter 5

Discussion

This chapter discusses the main findings, important design decisions, current progress, and limitations of the proposed system.

5.1 Novel Contributions of the Project

The main contribution of this project is the design of a privacy-preserving citizen reporting system for local governance. The system combines blockchain, smart contracts, IPFS, AI-based moderation, and a web-based DApp into one framework. A key design decision is the separation of real citizen identity from on-chain report records. Citizens are first verified through the simulated GovID service. After verification, the system uses ticket IDs and pseudonymous key pairs for report submission. This helps citizens report public issues without directly exposing their identity on the blockchain.

Another important contribution is the use of a backend relayer. The relayer submits blockchain transactions on behalf of citizens. This improves usability because citizens do not need to manage gas fees or directly interact with blockchain nodes. The system also uses IPFS for off-chain storage. Images and larger report content are stored outside the blockchain, while only CIDs and hashes are stored on-chain. This reduces blockchain storage overhead while preserving data integrity.

AI-assisted moderation is used before reports are stored permanently. This helps reduce spam, abusive content, and irrelevant submissions. The moderation service is designed using multiple oracle workers and an aggregator, which reduces dependence on a single moderation node.

5.2 Overall System Validity

The current system design supports the main objectives of the project. The permissioned blockchain provides a tamper-evident record of report metadata. Smart

contracts provide the base for report creation, state tracking, and future government actions.
The identity mechanism supports privacy by separating real identity verification from blockchain activity. The DApp improves accessibility by giving citizens a simple interface for report submission. The relayer hides blockchain complexity from users.
The AI oracle service has been deployed as a containerized service on a VPS. This shows that off-chain moderation can be integrated with the reporting workflow. IPFS is planned to handle media and report content storage.
At this progress stage, the system is valid as a proof-of-concept. However, full validation requires complete integration between the DApp, relayer, IPFS, smart contracts, and blockchain.

5.3 Achievement of Proposed Objectives

The project has made progress toward the proposed objectives.
The first objective was to design a permissioned blockchain using PoA consensus. The architecture has been designed around a private Ethereum-compatible blockchain. This supports integrity and controlled participation.
The second objective was to use smart contracts for reporting workflows. The smart contract design includes report creation, status management, and event handling. Government-side actions are still under development.
The third objective was to develop user-friendly interfaces. The citizen-facing DApp interfaces for login and report submission have been developed. Further work is needed for government-side interfaces.
The fourth objective was to integrate AI-based moderation. A text-based AI oracle moderation service has been implemented and deployed. Image moderation is planned as a future extension.
Overall, the project has achieved good progress at subsystem level. The next important task is full end-to-end integration.

5.4 Summary of Results and Observations

Several observations were made during the current progress stage.
First, the modular architecture helped the team develop different parts of the system in parallel. Each component can be tested separately before full integration.
Second, the hybrid storage approach is necessary. Blockchain is suitable for metadata and audit trails, but not for large files. Therefore, IPFS is required for storing media and report content.

Third, the relayer is important for usability. It allows citizens to use the system without understanding blockchain transactions or gas fees.
Fourth, AI moderation should happen before IPFS upload and blockchain submission. This reduces the risk of storing harmful or irrelevant content permanently.
Finally, containerized deployment improved service isolation and deployment consistency, especially for the oracle subsystem.

5.5 Real-world Applicability

The proposed system can be useful for local governments where citizens report issues such as road damage, garbage disposal, drainage problems, broken streetlights, and flooding.
In a real-world scenario, a citizen can submit a report through the DApp. The report is verified, moderated, stored off-chain, and recorded on the blockchain. Later, the citizen can check whether the report still exists and track its status.
This can improve trust between citizens and local authorities. It can also create an auditable record of how public issues are handled.

5.6 Limitations of the Current Prototype

The current prototype has some limitations.
The identity system is simulated. It does not implement a complete real-world zero-knowledge identity system.
The AI moderation service currently focuses mainly on text moderation. Image moderation is not fully implemented yet.
The oracle decentralization is simulated using separate containers. In a stronger real-world deployment, oracle nodes should run on separate servers or be operated by different organizations.
Full end-to-end integration is still in progress. Some components are implemented and tested separately, but the complete report submission flow must still be validated.
The system is also tested only in a controlled environment. Real-world deployment would require legal, security, and institutional approval.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This project presents a blockchain-based, community-assisted, privacy-preserving reporting framework for local governance. The system is designed to address common problems in centralized reporting platforms, such as lack of transparency, weak data integrity, privacy concerns, and unreliable report quality.
The proposed framework combines a permissioned blockchain, smart contracts, a simulated identity service, a backend relayer, IPFS, AI-assisted moderation, and a web-based DApp. Each component has a clear role in the overall system.
The blockchain and smart contracts provide tamper-evident report records and workflow management. The GovID simulator and ticket-based mechanism support privacy-preserving access control. The relayer improves usability by handling blockchain transactions on behalf of citizens. IPFS supports scalable off-chain storage. The AI oracle service helps filter spam and inappropriate content before permanent recording.
At the current progress stage, several important subsystems have been designed and partially implemented. The AI moderation subsystem has also been deployed as separate oracle containers on a VPS. This shows that the proposed architecture is technically feasible as a prototype.
The main insight from this project is that blockchain alone is not enough for a practical civic reporting system. It must be combined with privacy mechanisms, off-chain storage, moderation, and user-friendly interfaces.

6.2 Future Work

The next major task is to complete full end-to-end integration. The DApp, GovID simulator, backend relayer, AI oracle service, IPFS, smart contracts, and blockchain must work together in one complete report submission flow.

The government-side workflow should also be completed. This includes report review, issue assignment, resolution updates, closure, and citizen feedback.
Image moderation should be added to the AI moderation service. This will allow the system to check uploaded images for harmful, irrelevant, or misleading content.
The current identity system should be improved in future work. A real zero-knowledge proof or self-sovereign identity mechanism can replace the simulated GovID service.
The oracle system can also be improved. At present, decentralization is simulated using separate containers. In future, oracle nodes can be deployed on separate servers or operated by different trusted organizations.
More testing is required after integration. This includes blockchain transaction latency, IPFS upload time, oracle response time, and complete report submission time.
Security testing is also required before any real-world deployment. Smart contracts, APIs, private keys, relayer logic, and deployment infrastructure must be audited carefully.
Finally, the DApp should be improved through usability testing. Citizens should be able to submit and track reports easily without needing blockchain knowledge.

6.3 Final Remarks

The proposed system provides a strong foundation for a transparent and privacy-preserving local governance reporting platform. Although the prototype is still under development, the current progress shows that the architecture is feasible.
With further integration, testing, and improvement, this framework can support more trustworthy citizen participation and better accountability in local governance.

Bibliography

[1] M. Aakunuri, S. S. Kaatha, L. Battini, S. Jangam, and A. Thangella, "Decentralized complaint management system for law enforcement," *International Journal of Research Publication and Reviews (IJRPR)*, vol. 6, no. 6, pp. 9846–9854, 2025.

[2] R. Goyal and N. Mittal, "E-governance through blockchain technology. a review," in *2021 2nd Global Conference for Advancement in Technology (GCAT)*, pp. 1–4, 2021.

[3] T. Oliveira, M. Campolargo, J. Martins, and F. Cruz-Jesus, "Digital transformation in smart cities: The role of information and communication technologies," *Sustainability*, vol. 12, no. 9, pp. 1–20, 2020.

[4] J. Hamill, "Blockchain technology: Local government applications and challenges," white paper, International City/County Management Association and Government Finance Officers Association, oct 2018.

[5] Y. Goswami, A. Agrawal, and A. Bhatia, "E-governance: A tendering framework using blockchain with active participation of citizens," in *2020 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*, pp. 1–4, IEEE, 2020.

[6] L. H. Pratheeba, D. R. Bharath, N. E. Cibiya, S. Dheekshitha, and M. N. Divya, "Blockchain-based system for effective police complaint management," *Journal of Emerging Technologies and Innovative Research (JETIR)*, vol. 10, no. 6, 2024. June 2023. Available at `https://www.jetir.org`.

[7] "Decision analytics using permissioned blockchain "commledger"." UND Scholarly Commons, 2026. Accessed on January 7, 2026.

[8] ResearchGate, "A survey on blockchain consensus with a performance comparison of pow, pos and pure pos," *ResearchGate*, 2020.

[9] arxiv, "Performance modeling of public permissionless blockchains: A survey," *arXiv preprint arXiv:2402.18049*, 2024.

[10] ResearchGate, "A survey on blockchain consensus with a performance comparison of pow, pos and pure pos," *ResearchGate*, 2020.

[11] "What is proof of authority: A beginner's guide." ZebPay Blog, 2026. Accessed on January 7, 2026.

[12] "Proof of authority blockchain: Private lightweight, fast consensus." Echo, 2026. Accessed on January 7, 2026.

[13] "Proof of authority." Grokipedia, 2026. Accessed on January 7, 2026.

[14] "Beginner's guide to proof-of-authority (poa)." OneKey, 2026. Accessed on January 7, 2026.

[15] H. Journals, "The governance technology for blockchain systems: a survey," *Frontiers of Computer Science*, 2023.

[16] MDPI, "Unleashing the potential of permissioned blockchain: Addressing privacy, security, and interoperability concerns in healthcare data management," *Electronics*, vol. 13, no. 24, p. 5050, 2023.

[17] J. R. Andrey Nechesov, "Empowering government efficiency through civic intelligence," *MDPI*, 2023.

[18] M. Muneeb, Z. Raza, I. U. Haq, and O. Shafiq, "Smartcon: A blockchain-based framework for smart contracts and transaction management," *IEEE Access*, vol. 10, pp. 23687–23699, 2022.

[19] K. S. Pillai, "Smart contracts offer a toolkit to transform governance," 2023.

[20] A. CERTs, "The future of governance: Will blockchain revolutionize public administration?," 2023.

[21] M. Shuaib, S. Daud, S. Alam, and W. Khan, "Blockchain-based framework for secure and reliable land registry system," *TELKOMNIKA (Telecommunication Computing Electronics and Control)*, vol. 18, pp. 2560–2571, 10 2020.

[22] U. E. C. Johannes Pandeni Paavo, Rafael Rodriguez-Puentes, "Practicality of blockchain technology for land registration: A namibian case study," *MDPI*, 2023.

[23] W. Liang, "Scalable blockchain e-voting with off-chain computation and on-chain verification," *ResearchGate*, 2024.

[24] H. Baniata and G. Caluna, "Bp-vot: Blockchain-based e-voting using smart contracts, differential privacy, and self-sovereign identities," *IEEE Access*, vol. 13, pp. 46106–46123, 2025.

[25] N. Dimitri, "Quadratic voting in blockchain governance," *Information*, vol. 13, p. 305, 06 2022.

[26] P. N. M.M. Ibrahimy, A. Norta and H. Nowandish, "Transforming e-participatory budgeting with blockchain: Boosting transparency and citizen engagement," *IEEE Transactions on Engineering Management*, vol. 72, pp. 1376–1403, 2022.

[27] S. Park, M. Specter, N. Narula, and R. L. Rivest, "Going from bad to worse: from internet voting to blockchain voting," *Journal of Cybersecurity*, vol. 7, p. tyaa025, 01 2021.

[28] MDPI, "The development of user-centric design guidelines for web3 applications: An empirical study," 2025. Accessed on January 8, 2026.

[29] N. Sonule, "User experience (ux) in web3 and dapps: Challenges and opportunities," 2025. Accessed on January 8, 2026.

[30] Ramotion, "Ui/ux design for web 3.0," 2026. Accessed on January 8, 2026.

[31] F. Weekly, "Why usability is the missing layer in crypto finance," 2025. Accessed on January 8, 2026.

[32] O. Academic, "Governance and societal impact of blockchain-based self-sovereign identities," 2022. Accessed on January 8, 2026.

[33] Fraunhofer, "Conducting a usability evaluation of decentralized identity management solutions," 2026. Accessed on January 8, 2026.

[34] DZone, "Blockchain + ai integration: The architecture nobody's talking about," 2026. Accessed on January 8, 2026.

[35] C. V. Marian, D. A. Mitrea, D. S. Rusu, and A. Vasilateanu, "Transparent digital governance: A blockchain-based workflow audit application," 2025. Accessed on January 9, 2026.

[36] S. F. King and P. Brown, "Fix my street or else: Using the internet to voice local public service concerns," 2012. Accessed on January 8, 2026.

[37] mySociety Research, "Fixmystreet!," 2026. Accessed on January 9, 2026.

[38] O. Adeyeye, “BLOCKCHAIN AND THE FUTURE OF E-GOVERNMENT: STRATEGIC IMPLICATIONS FOR POLICYMAKERS,” 2025. Accessed on January 9, 2026.

[39] S. Nikonenko, “Blockchain UX explained: Challenges, trends, and real solutions,” 2025. Accessed on January 9, 2026.

[40] A. Bertrand and E. G. Consulting, “How can digital government connect citizens without leaving the disconnected behind?,” tech. rep., Ernst & Young (EY), June 2023. EY Connected Citizens Report. Accessed on January 9, 2026.

[41] N. Shahin and L. Ismail, “Towards trustworthy sign language translation system: A privacy-preserving edge–cloud–blockchain approach,” *Mathematics*, vol. 13, no. 23, p. 3759, 2025. Accessed on January 9, 2026.

[42] Tencent Cloud, “How can content moderation use blockchain technology?,” 2025. Accessed on January 9, 2026.

[43] M. A. Wani, M. ElAffendi, and K. A. Shakil, “AI-generated spam review detection framework with deep learning algorithms and natural language processing,” *Computers*, vol. 13, no. 10, p. 264, 2024. Accessed on January 9, 2026.

[44] G. Gosztonyi, D. Gyetván, and A. Kovács, “Theory and practice of social media’s content moderation by artificial intelligence in light of european union’s AI act and digital services act,” *Politics in Central Europe*, 2024. Accessed on January 9, 2026.

[45] V. Kolluri, T. R. Gatla, and S. T. Boppiniti, “AI-powered citizen science: Crowd-learning, data verification, and public engagement,” *Scientific Endeavours*, vol. 1, no. 1, pp. 53–60, 2025. Accessed on January 9, 2026.

[46] S. Mohammadi and T. Yasseri, “AI feedback enhances community-based content moderation through engagement with counterarguments,” October 6 2025. arXiv preprint arXiv:2507.08110v3. Accessed on January 9, 2026.

[47] S. Goundar and I. Gondal, “AI-blockchain integration for real-time cybersecurity: System design and evaluation,” *Cybersecurity and Privacy*, vol. 5, p. 59, August 14 2025. Accessed on January 9, 2026.

[48] N. Kader, I. Kang, and O. Seneviratne, “Enhancing web spam detection through a blockchain-enabled crowdsourcing mechanism,” October 1 2024. arXiv preprint arXiv:2410.00860v1. Accessed on January 9, 2026.

[49] J. Li, L. Jiang, H. Liang, T. Peng, S. Wang, and H. Wei, "Block-CITE: A blockchain-based crowdsourcing interactive trust evaluation," *Applied System Innovation*, vol. 6, p. 245, October 1 2025. Accessed on January 9, 2026.

[50] Kava News, "AI-powered oracles: Bridging blockchains with smarter data," May 29 2024. Accessed on January 9, 2026.

[51] K. Zintus-art, B. Vass, and J. Ward, "Empirical evidence in AI oracle development," March 21 2025. Chainlink Blog. Accessed on January 9, 2026.

[52] CodeZeros, "AI-powered oracles: How generative AI is enhancing blockchain data feeds," April 10 2025. Accessed on January 9, 2026.

[53] M. Prasad, "Blockchain + AI in combating deepfake content circulation," *Scientific Journal of Artificial Intelligence and Blockchain Technologies*, April 2024. Accessed on January 9, 2026.

[54] A. Kumar, "Hybrid moderation models: Balancing AI and human oversight," October 27 2025. Infosys BPM. Accessed on January 9, 2026.

[55] B. Kae, "Research: AI agent sector overview," March 10 2025. The Block Research. Accessed on January 9, 2026.

[56] Chase Advisors, "Ai and content moderation," tech. rep., Chase Advisors, January 2024. Accessed on January 9, 2026.

[57] K. Mounika and N. R. Reddy, "An integrated machine learning framework for spammer and fake user detection in online social networks," *Fringe Management, Economics and Politics*, 2024. Accessed on January 9, 2026.